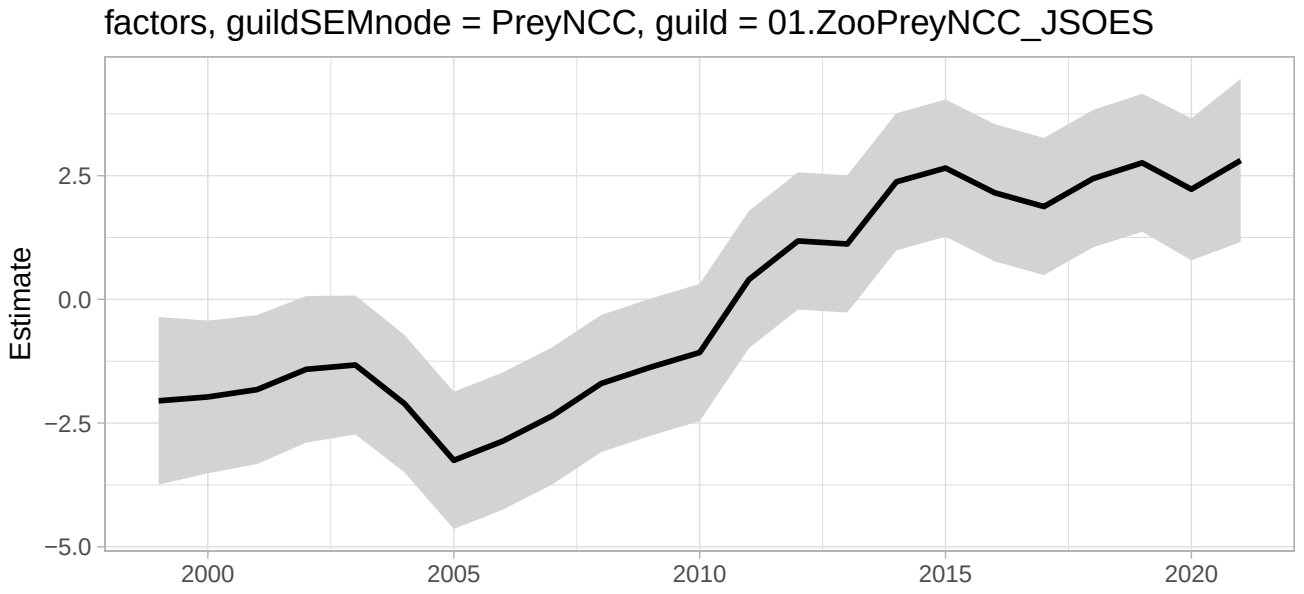
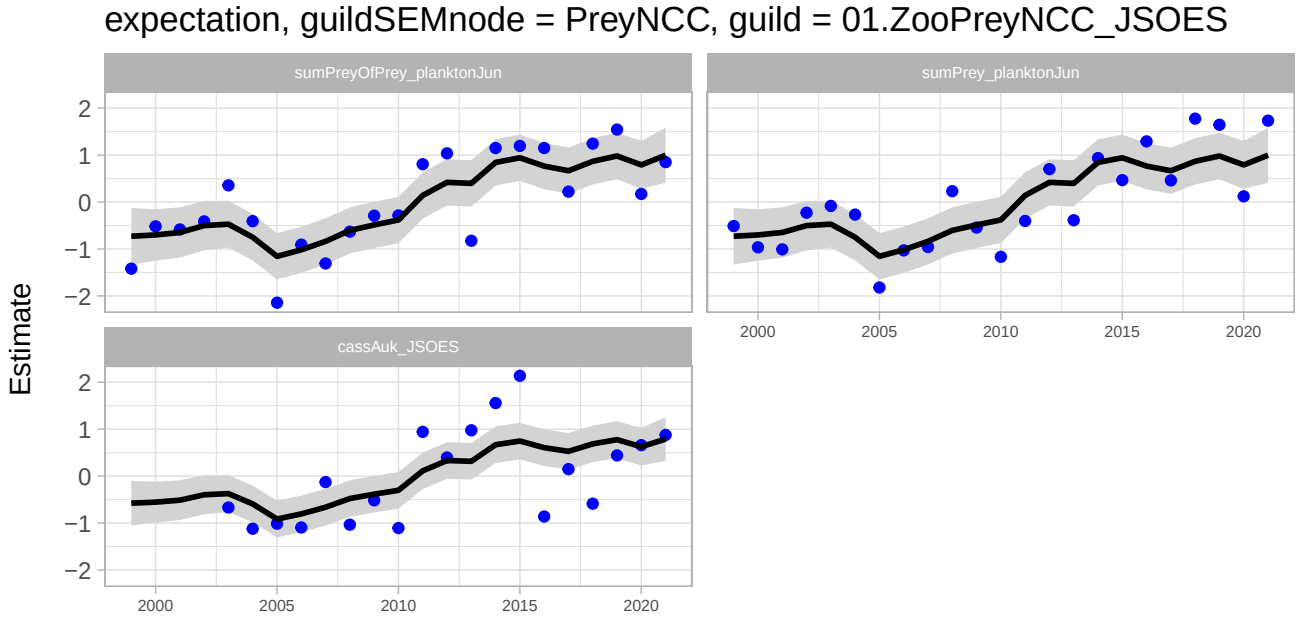
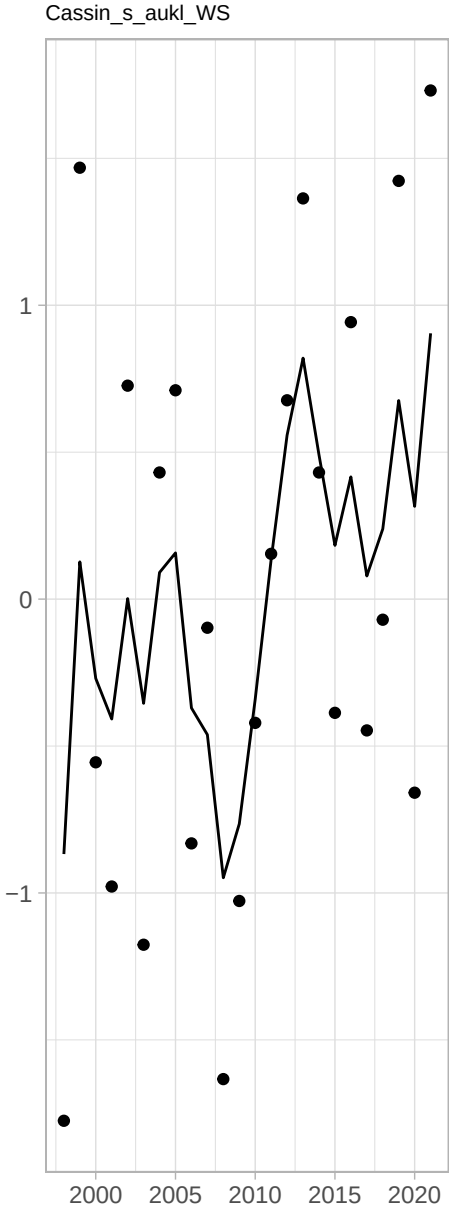
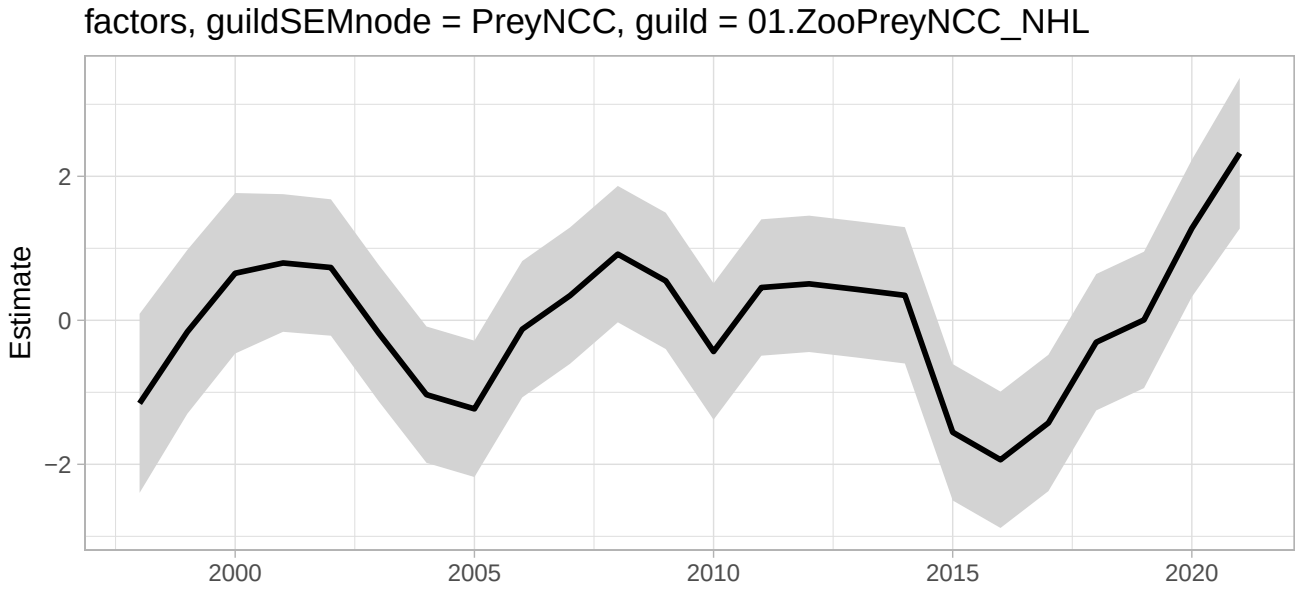
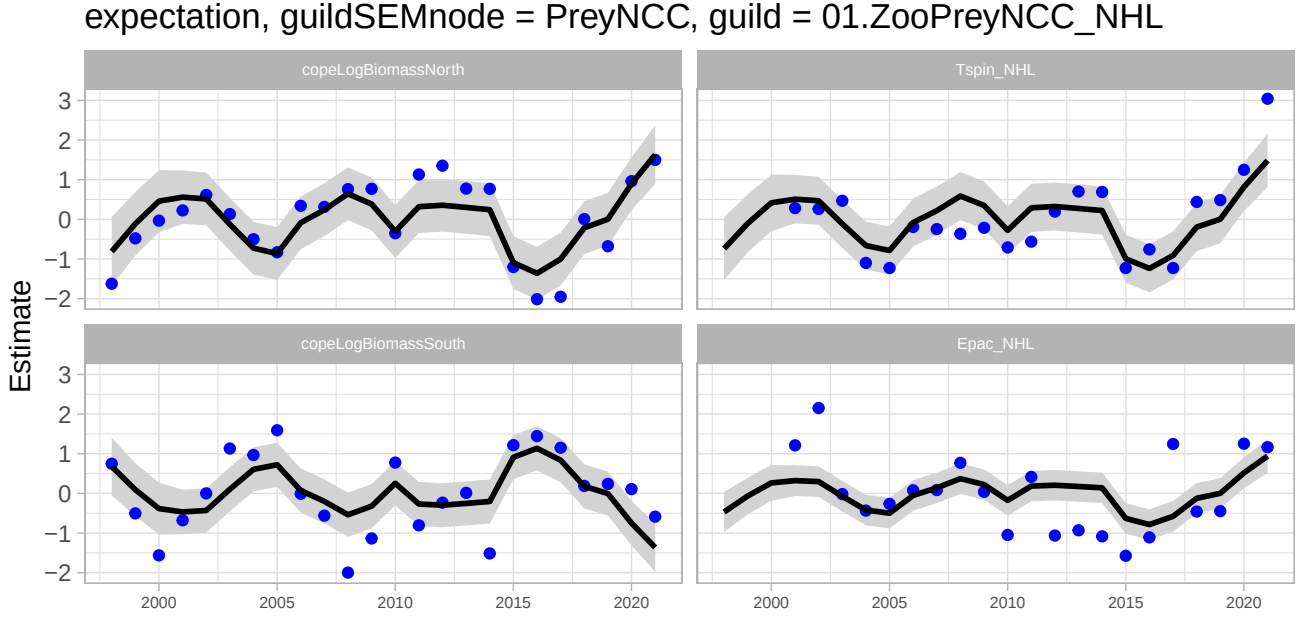

factors, guildSEMnode = PreyNCC, guild = 01.ZooPreyNCC_JSOES
Estimate
2.5
0.0
−2.5
−5.0
2000
2005
2010
2015
2020

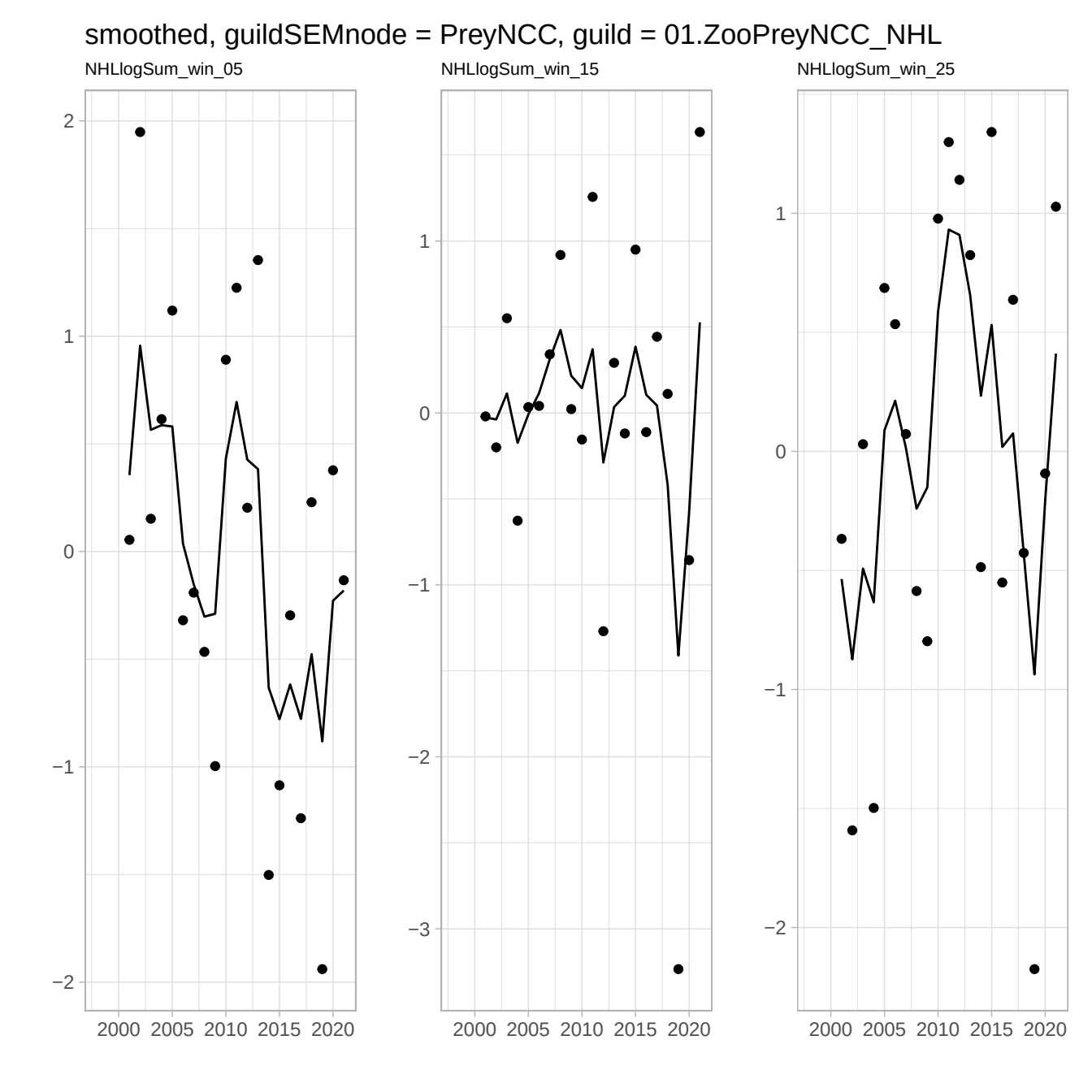

expectation, guildSEMnode = PreyNCC, guild = 01.ZooPreyNCC_JSOES
Estimate
sumPreyOfPrey_planktonJun
sumPrey_planktonJun
cassAuk_JSOES
2
1
0
−1
−2
2000
2005
2010
2015
2020

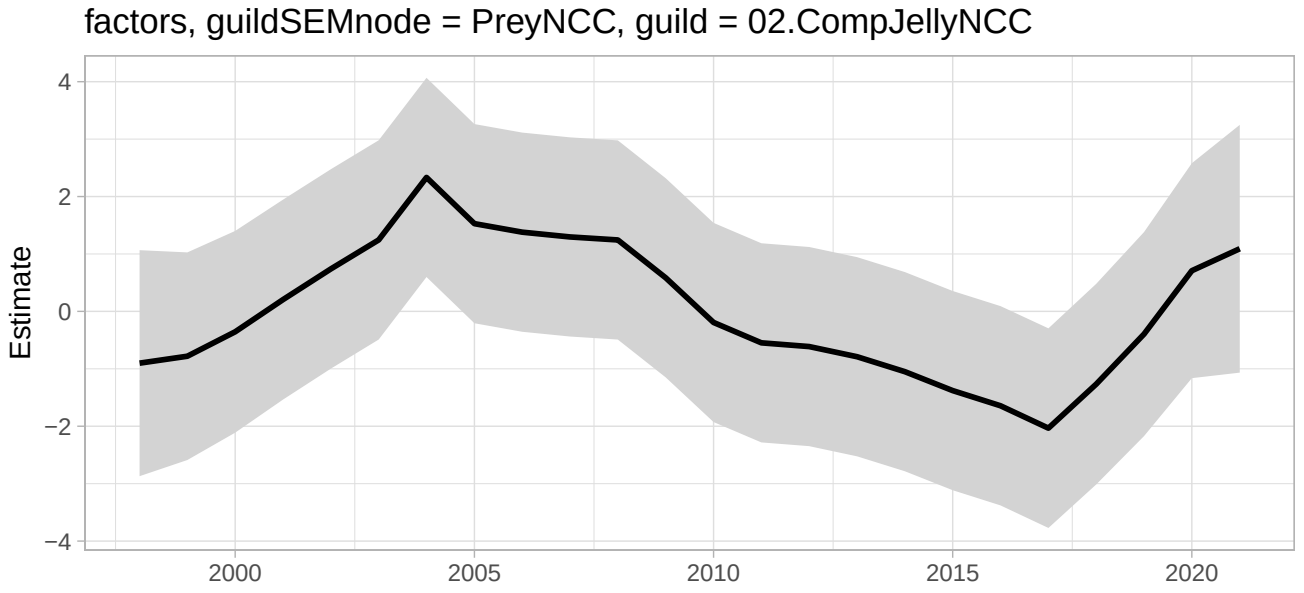
smoothed, guildSEMnode = PreyNCC, guild = 01.ZooPreyNCC_JSOES

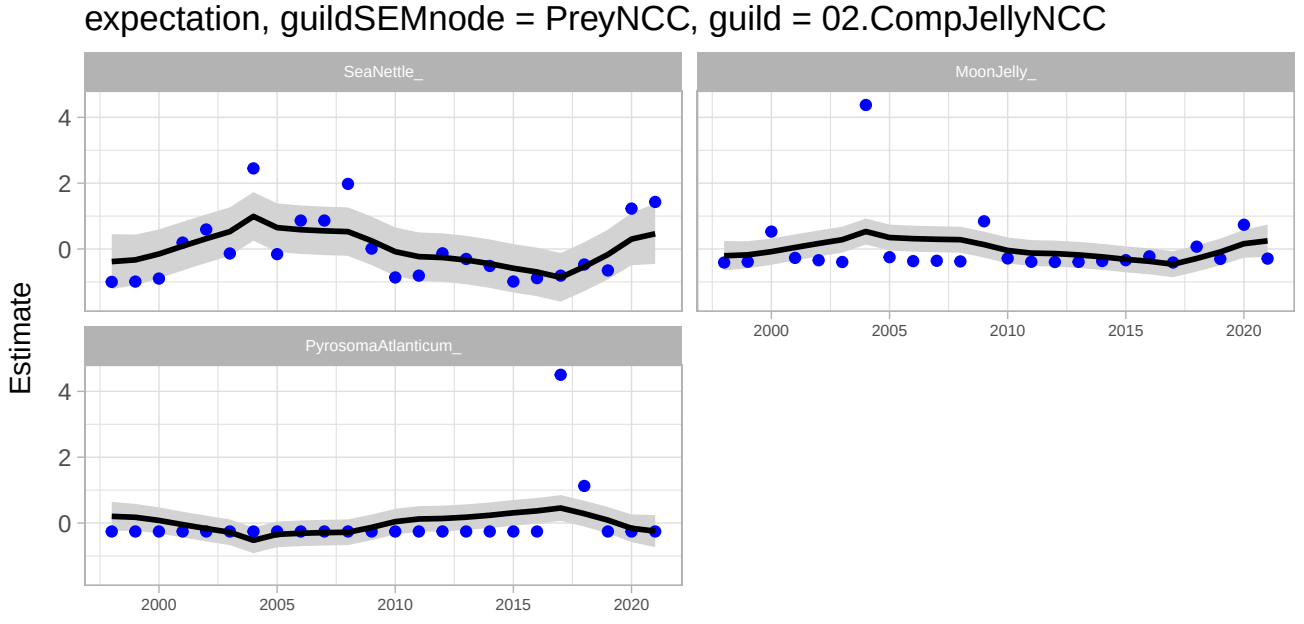


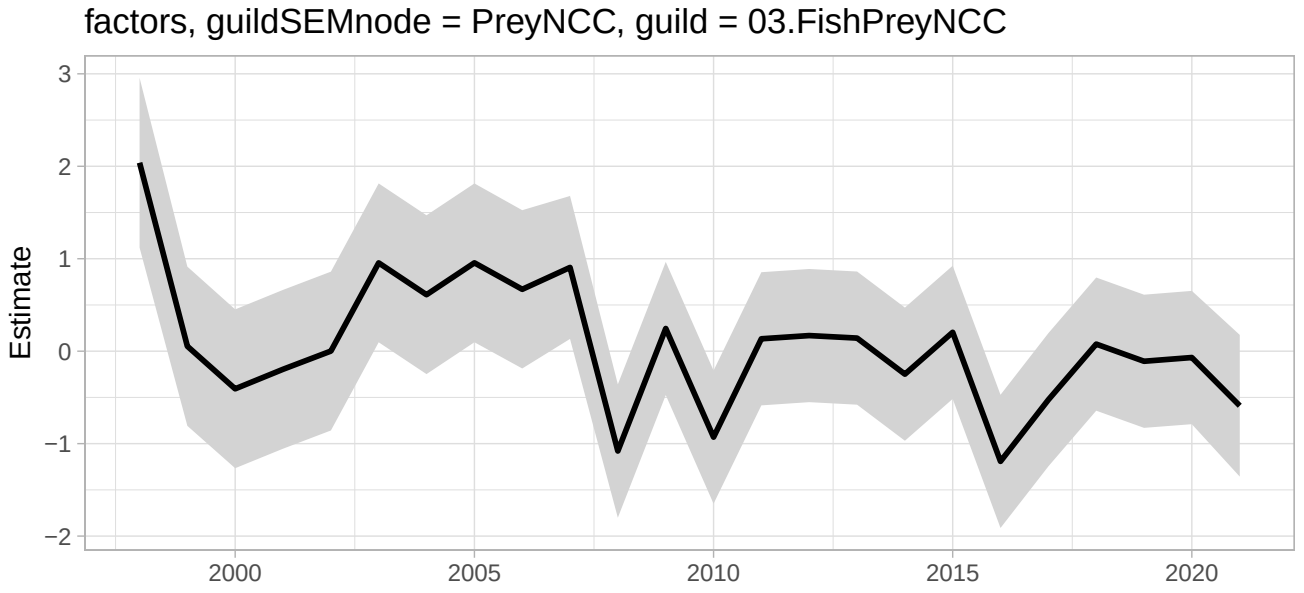

factors, guildSEMnode = PreyNCC, guild = 01.ZooPreyNCC_NHL
Estimate
2
0
−2
2000
2005
2010
2015
2020

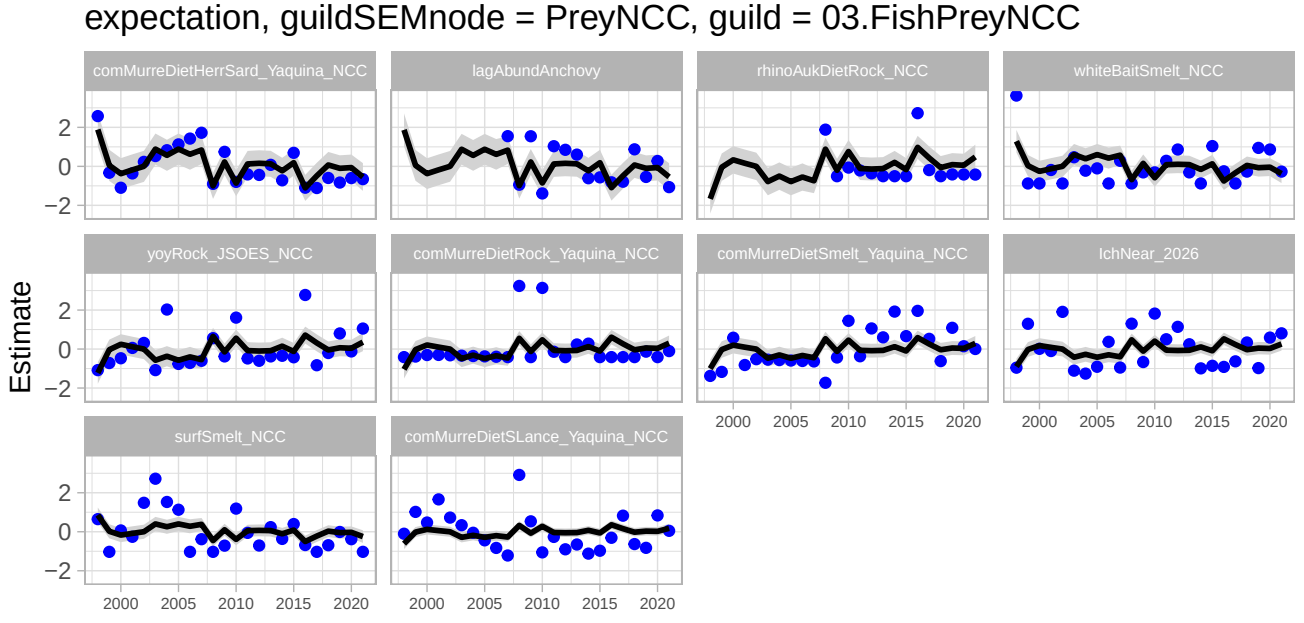

expectation, guildSEMnode = PreyNCC, guild = 01.ZooPreyNCC_NHL
Estimate
copeLogBiomassNorth
Tspin_NHL
copeLogBiomassSouth
Epac_NHL
3
2
1
0
−1
−2
2000
2005
2010
2015
2020


smoothed, guildSEMnode = PreyNCC, guild = 01.ZooPreyNCC_NHL
NHLlogSum_win_05
NHLlogSum_win_15
NHLlogSum_win_25
2
1
0
−1
−2
−3
2000
2005
2010
2015
2020

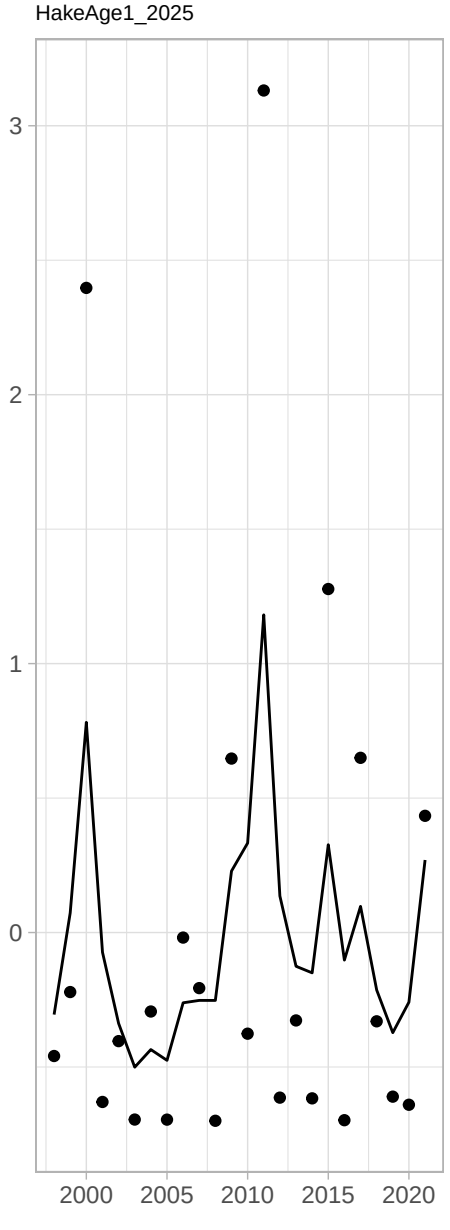

factors, guildSEMnode = PreyNCC, guild = 02.CompJellyNCC
Estimate
4
2
0
−2
−4
2000
2005
2010
2015
2020


expectation, guildSEMnode = PreyNCC, guild = 02.CompJellyNCC
SeaNettle_
MoonJelly_
PyrosomaAtlanticum_
Estimate
4
2
0
2000
2005
2010
2015
2020


factors, guildSEMnode = PreyNCC, guild = 03.FishPreyNCC
Estimate
3
2
1
0
−1
−2
2000
2005
2010
2015
2020

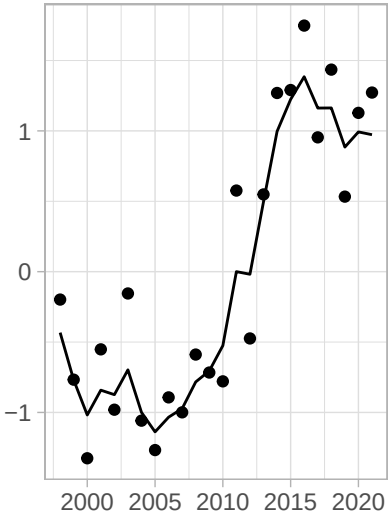

expectation, guildSEMnode = PreyNCC, guild = 03.FishPreyNCC
Estimate
comMurreDietHerrSard_Yaquina_NCC
lagAbundAnchovy
rhinoAukDietRock_NCC
whiteBaitSmelt_NCC
yoyRock_JSOES_NCC
comMurreDietRock_Yaquina_NCC
comMurreDietSmelt_Yaquina_NCC
IchNear_2026
surfSmelt_NCC
comMurreDietSLance_Yaquina_NCC
2
0
−2
2000
2005
2010
2015
2020

smoothed, guildSEMnode = PreyNCC, guild = 03.FishPreyNCC

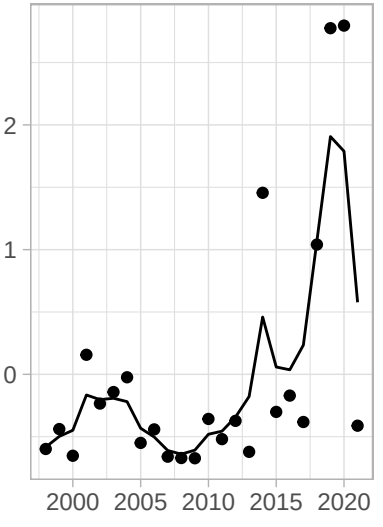


smoothed, guildSEMnode = PreyNCC, guild = 04.CompFishNCC

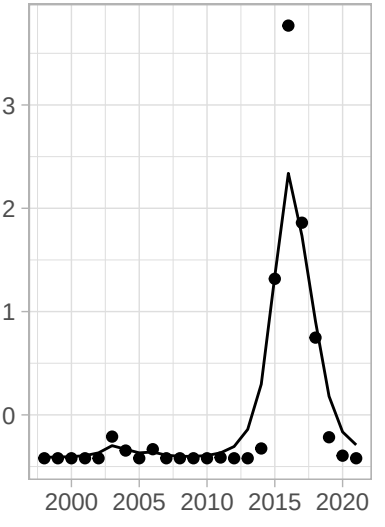
marketsquid_GAM



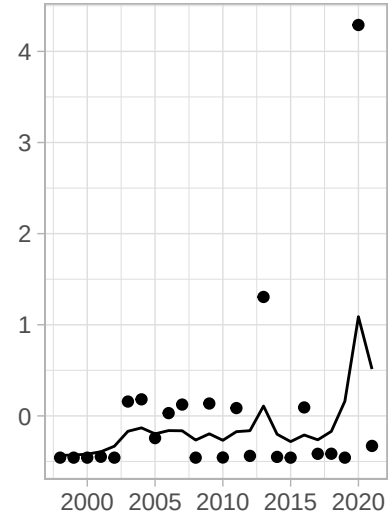
CaliforniaMarketSquid_

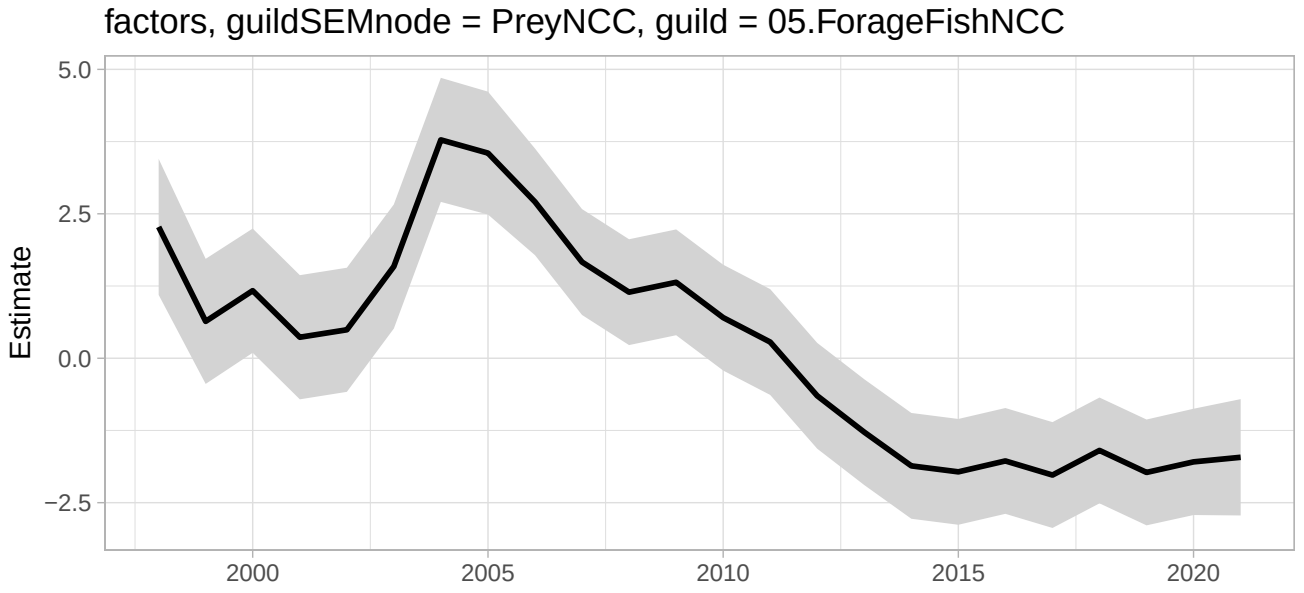


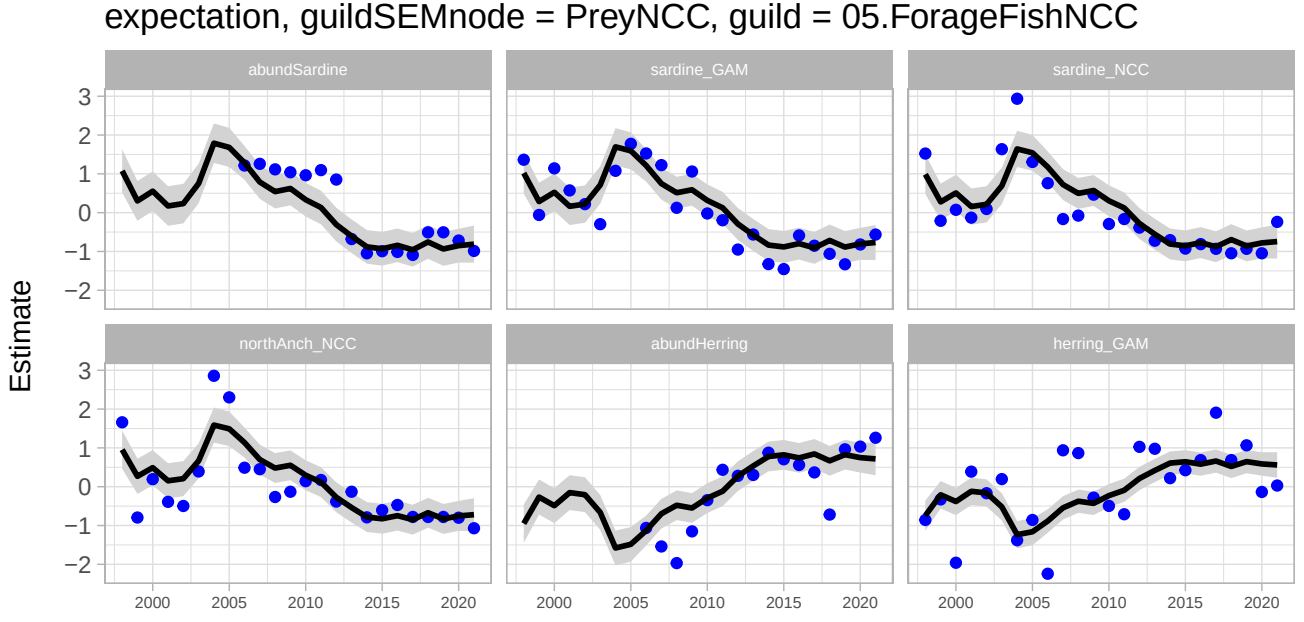
PacificPompano_



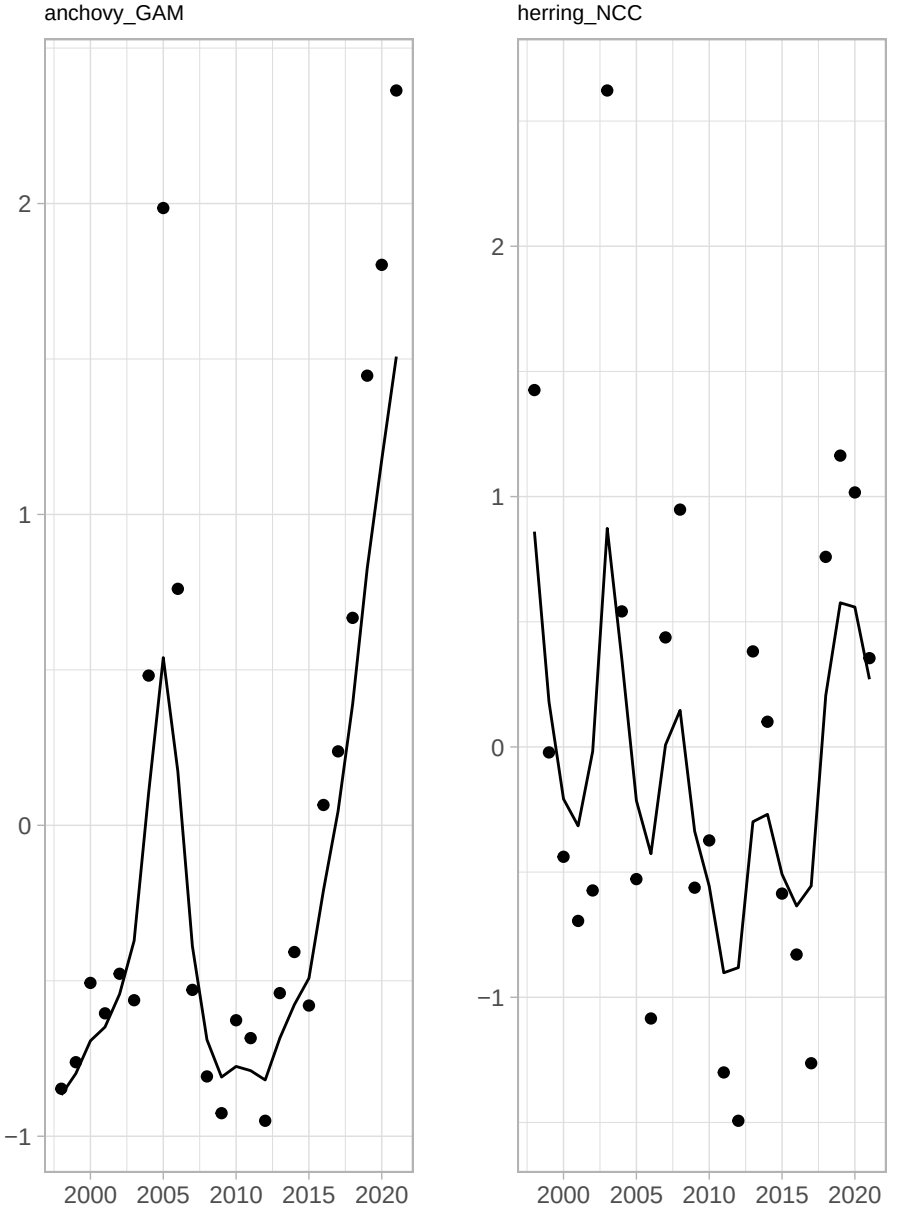
Sablefish__JSOES

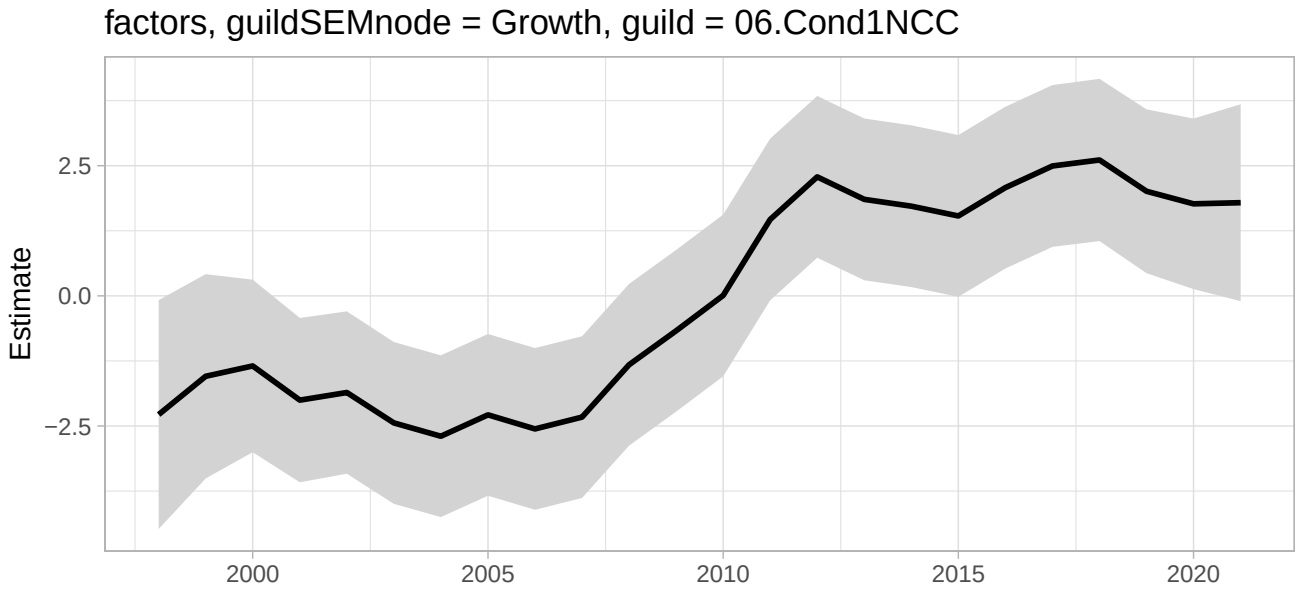


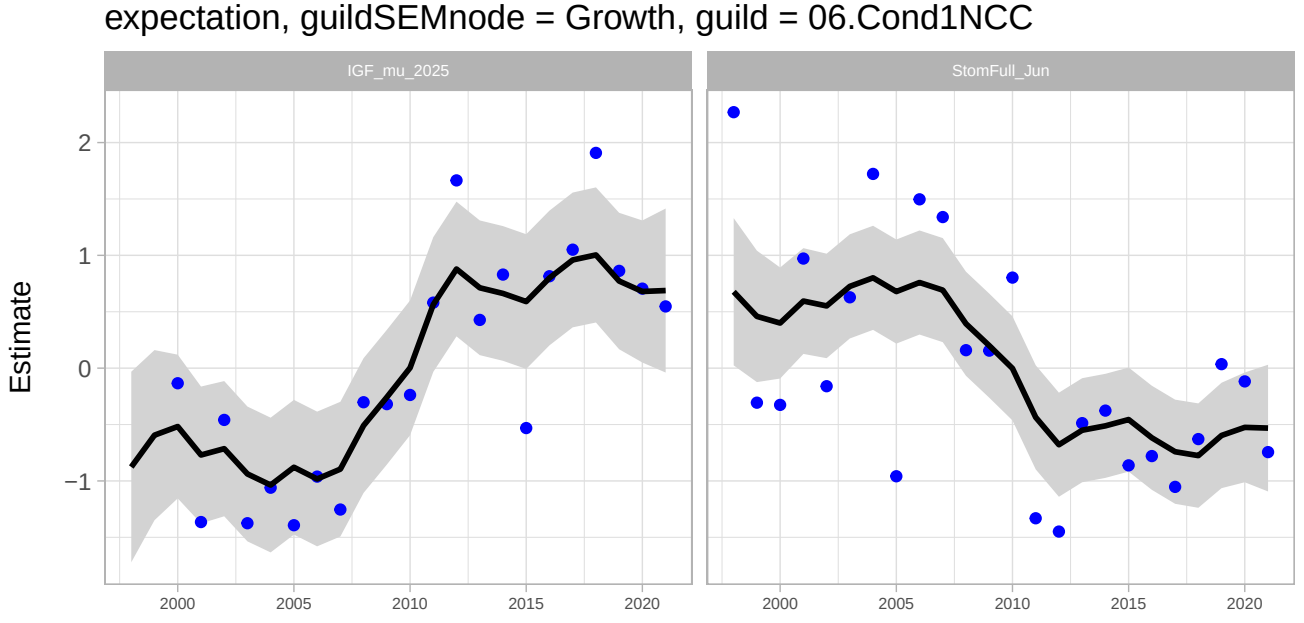

factors, guildSEMnode = PreyNCC, guild = 05.ForageFishNCC
Estimate
5.0
2.5
0.0
−2.5
2000
2005
2010
2015
2020

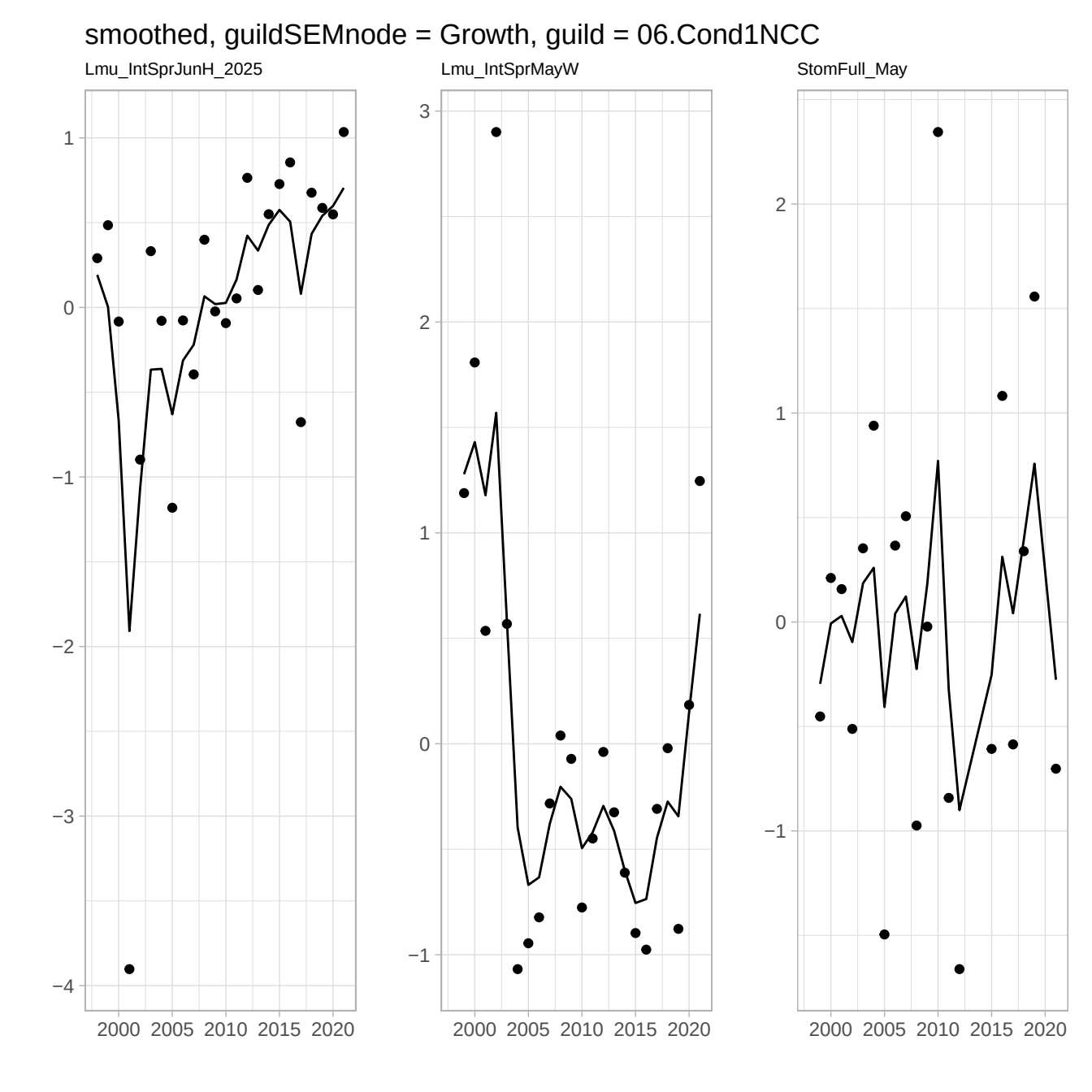

expectation, guildSEMnode = PreyNCC, guild = 05.ForageFishNCC
Estimate
abundSardine
sardine_GAM
sardine_NCC
northAnch_NCC
abundHerring
herring_GAM
3
2
1
0
−1
−2
2000
2005
2010
2015
2020

smoothed, guildSEMnode = PreyNCC, guild = 05.ForageFishNCC

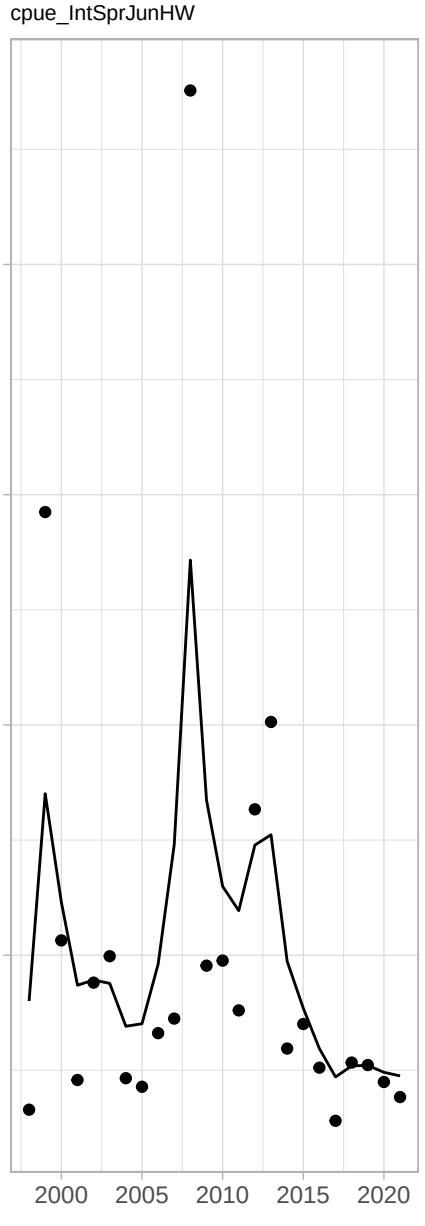


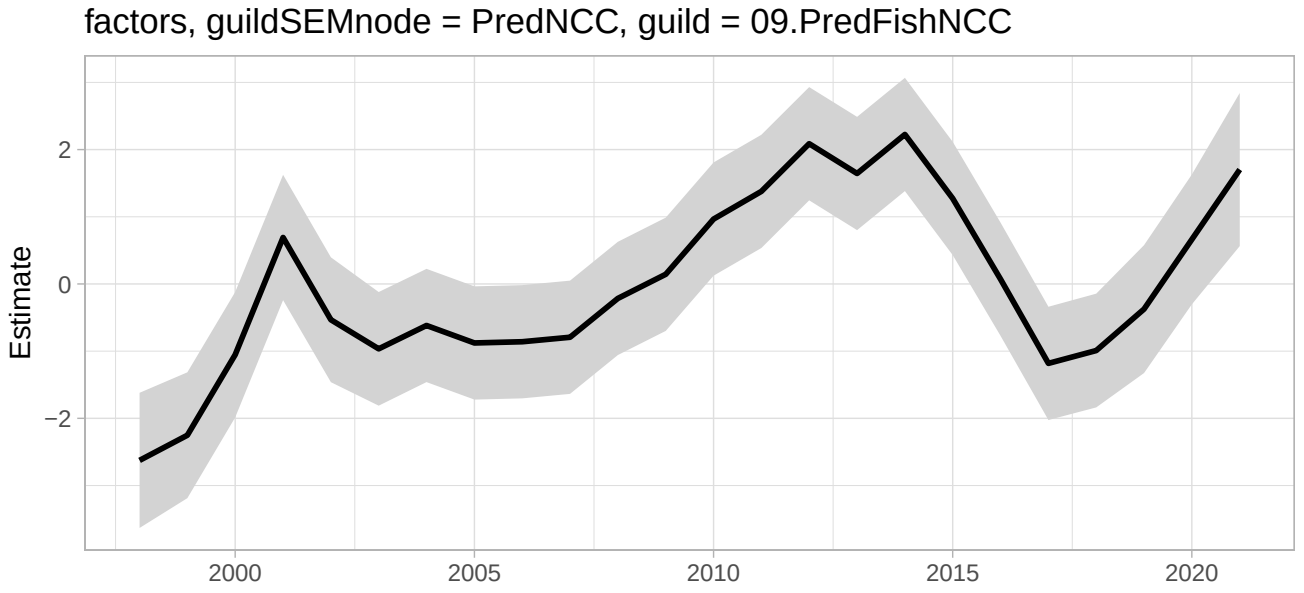

factors, guildSEMnode = Growth, guild = 06.Cond1NCC
Estimate
2.5
0.0
−2.5
2000
2005
2010
2015
2020

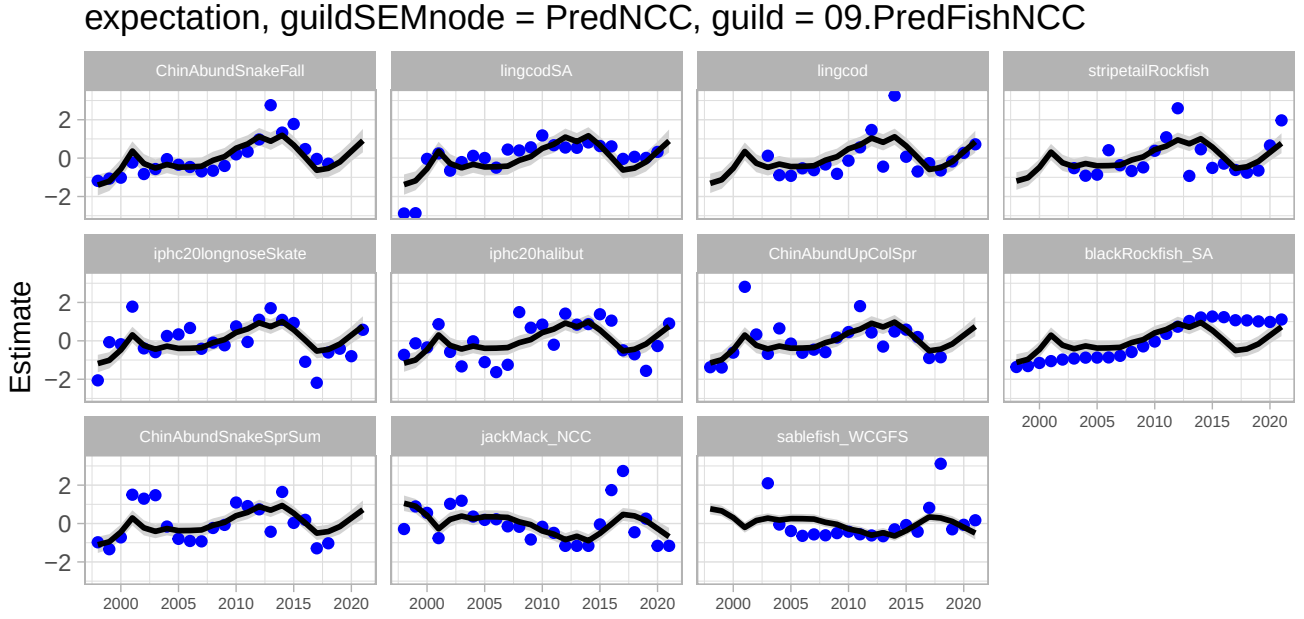

expectation, guildSEMnode = Growth, guild = 06.Cond1NCC
IGF_mu_2025
StomFull_Jun
Estimate
2
1
0
−1
2000
2005
2010
2015
2020

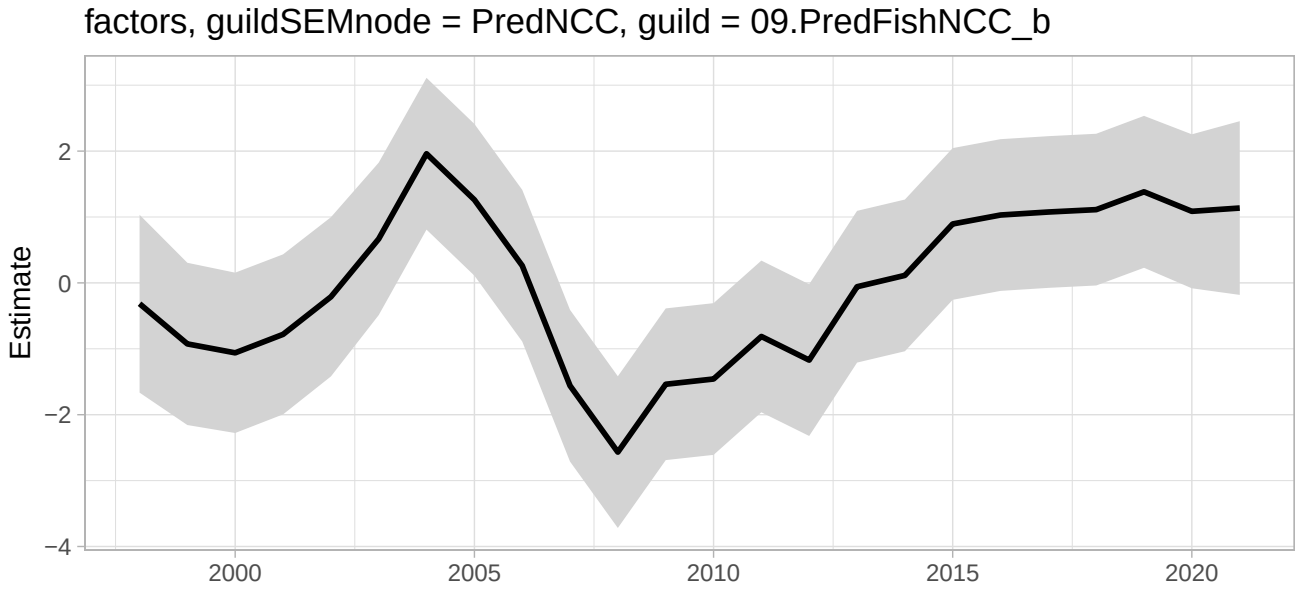

smoothed, guildSEMnode = Growth, guild = 06.Cond1NCC
Lmu_IntSprJunH_2025
Lmu_IntSprMayW
StomFull_May
1
0
−1
−2
−3
−4
3
2
1
0
−1
2
1
0
−1
2000
2005
2010
2015
2020

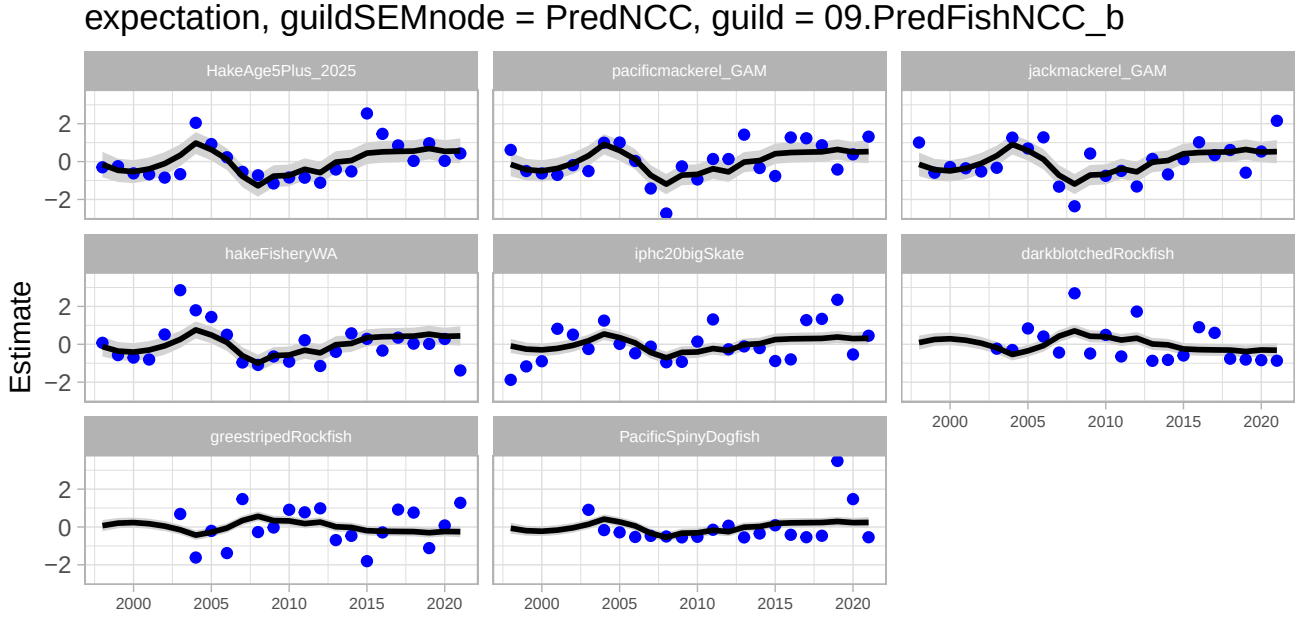
smoothed, guildSEMnode = Abundance, guild = 07.Cond2NCC

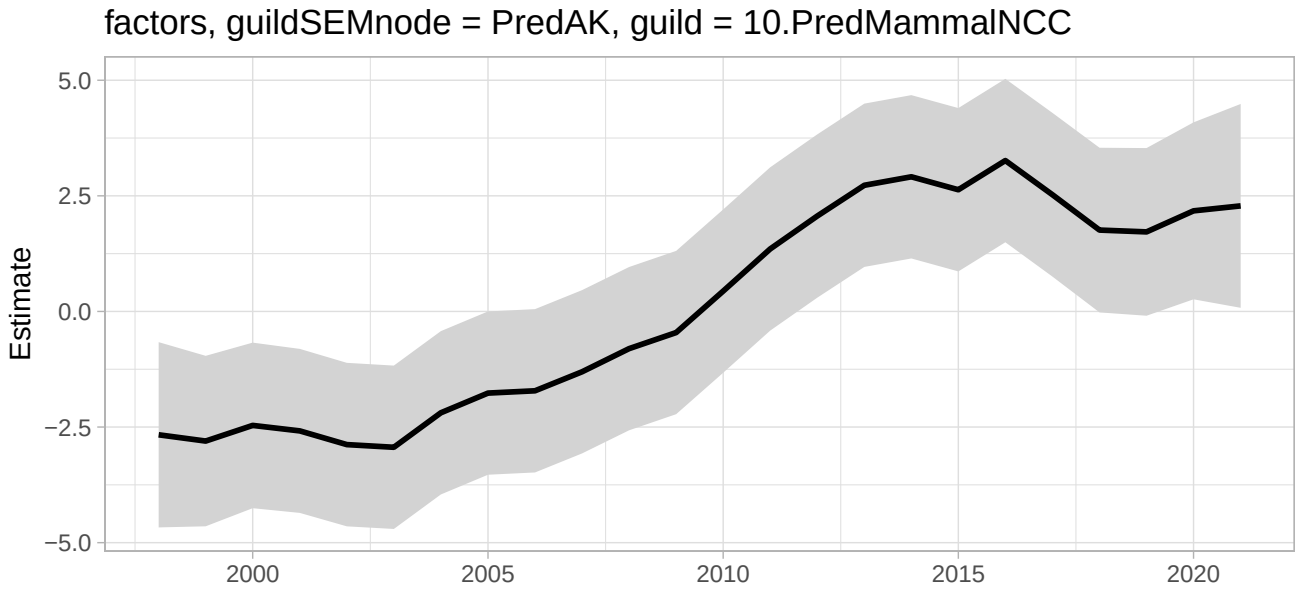


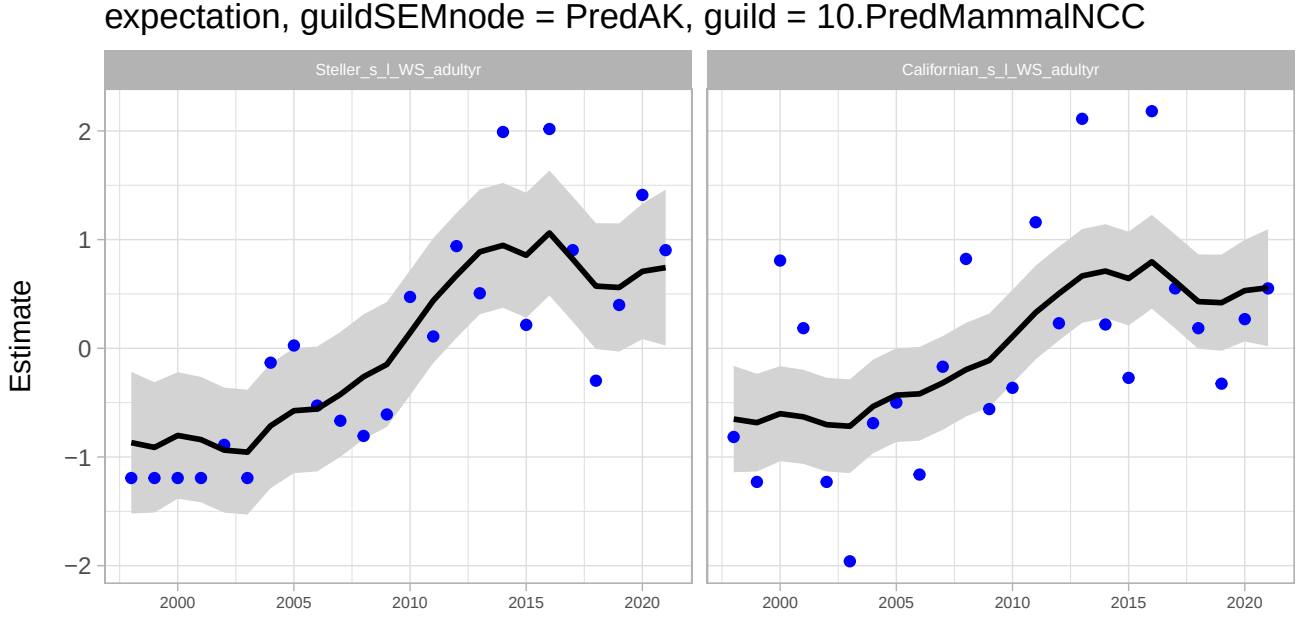

factors, guildSEMnode = PredNCC, guild = 09.PredFishNCC
Estimate
2
0
−2
2000
2005
2010
2015
2020


expectation, guildSEMnode = PredNCC, guild = 09.PredFishNCC
Estimate
ChinAbundSnakeFall
lingcodSA
lingcod
stripetailRockfish
iphc20longnoseSkate
iphc20halibut
ChinAbundUpColSpr
blackRockfish_SA
ChinAbundSnakeSprSum
jackMack_NCC
sablefish_WCGFS
2
0
−2
2000
2005
2010
2015
2020

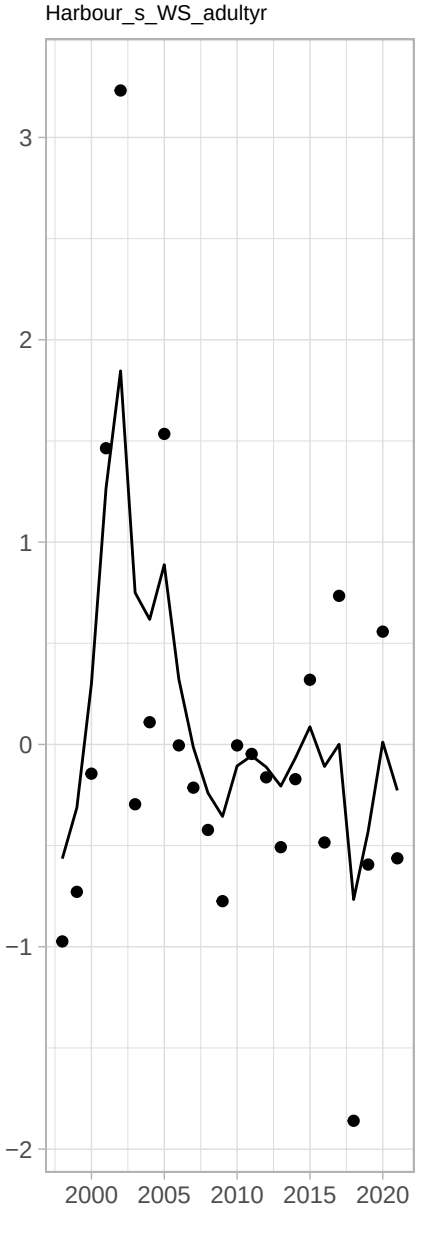

factors, guildSEMnode = PredNCC, guild = 09.PredFishNCC_b
Estimate
2
0
−2
−4
2000
2005
2010
2015
2020

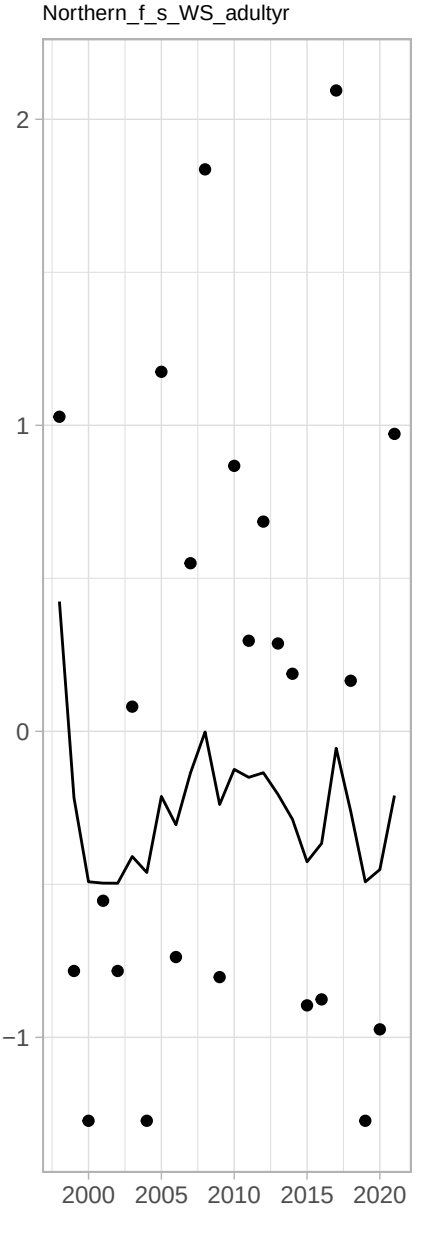

expectation, guildSEMnode = PredNCC, guild = 09.PredFishNCC_b
Estimate
HakeAge5Plus_2025
pacificmackerel_GAM
jackmackerel_GAM
hakeFisheryWA
iphc20bigSkate
darkblotchedRockfish
greestripedRockfish
PacificSpinyDogfish
2
0
−2
2000
2005
2010
2015
2020


factors, guildSEMnode = PredAK, guild = 10.PredMammalNCC
Estimate
5.0
2.5
0.0
−2.5
−5.0
2000
2005
2010
2015
2020


expectation, guildSEMnode = PredAK, guild = 10.PredMammalNCC
Steller_s_l_WS_adultyr
Californian_s_l_WS_adultyr
Estimate
2
1
0
−1
−2
2000
2005
2010
2015
2020

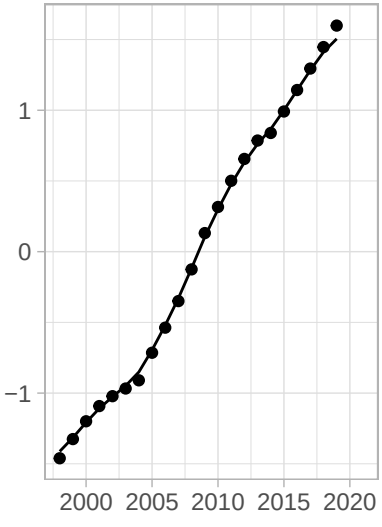
smoothed, guildSEMnode = PredAK, guild = 10.PredMammalNCC



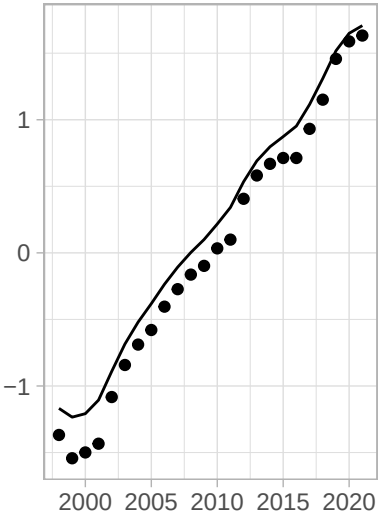


smoothed, guildSEMnode = PredAK, guild = 10.PredMammalNCC_BCAK

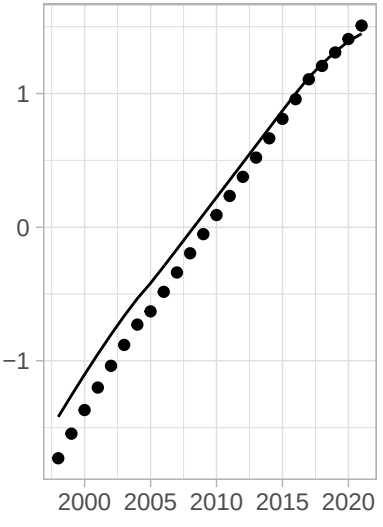
ssl_est_wholerange_adultyr



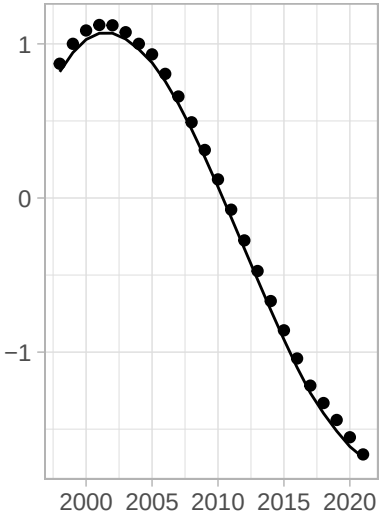
Killer_whales_NR_BC_adultyr

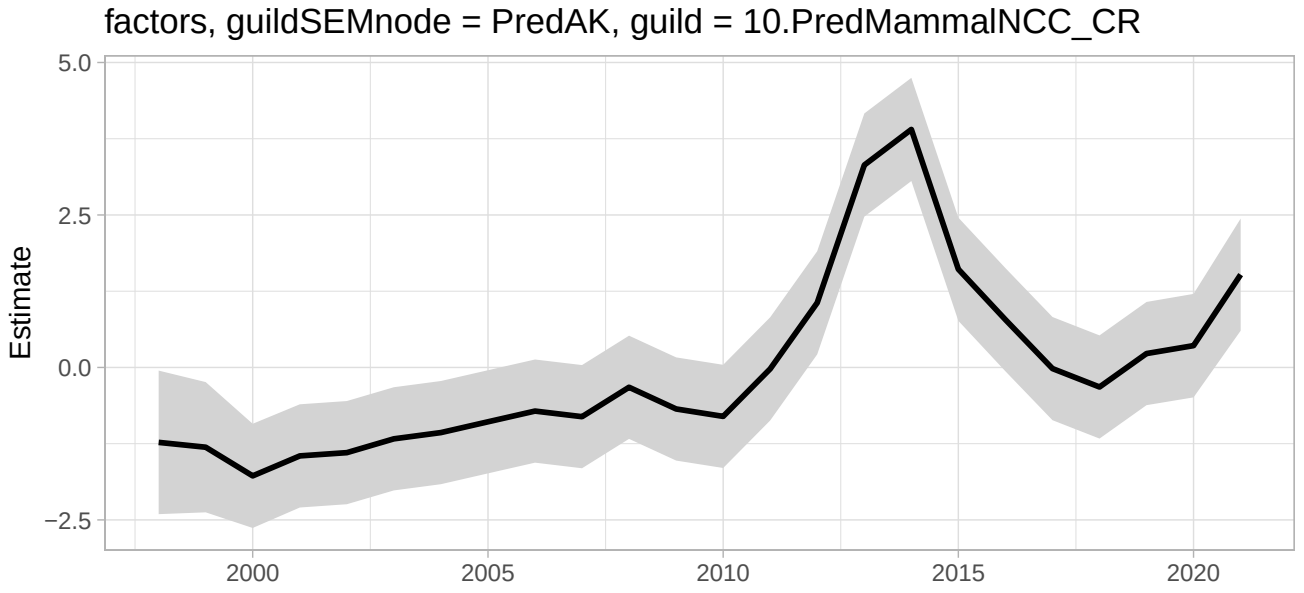


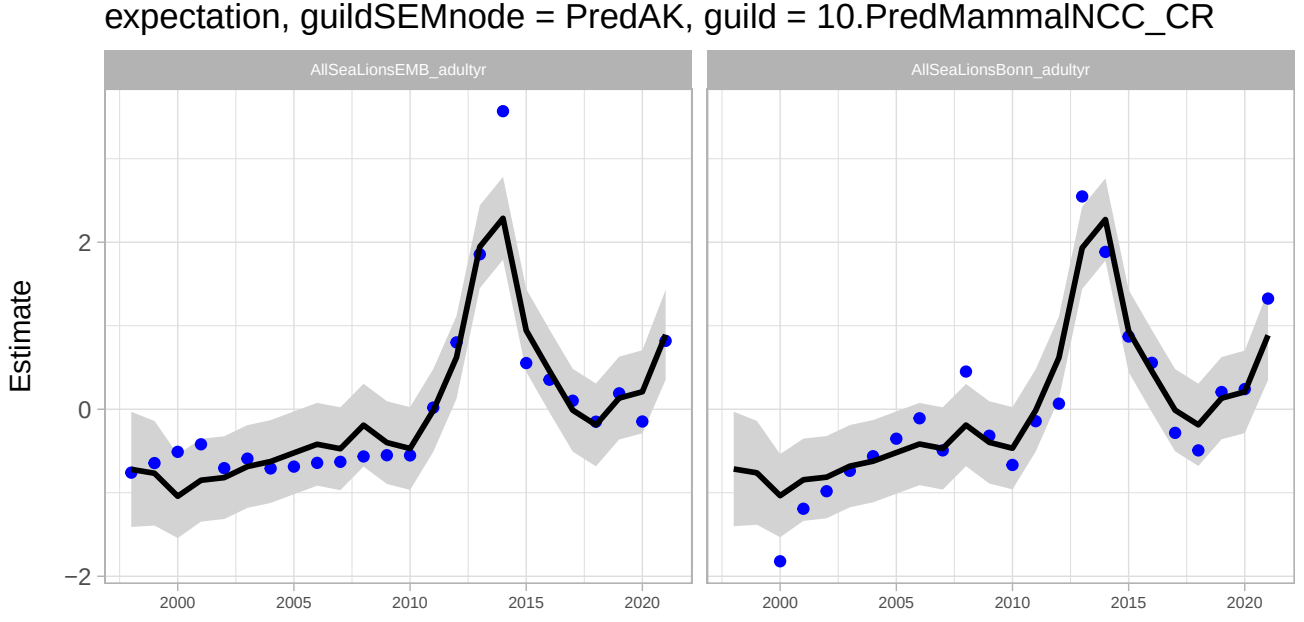
harbour_seal_DFO_WCVI

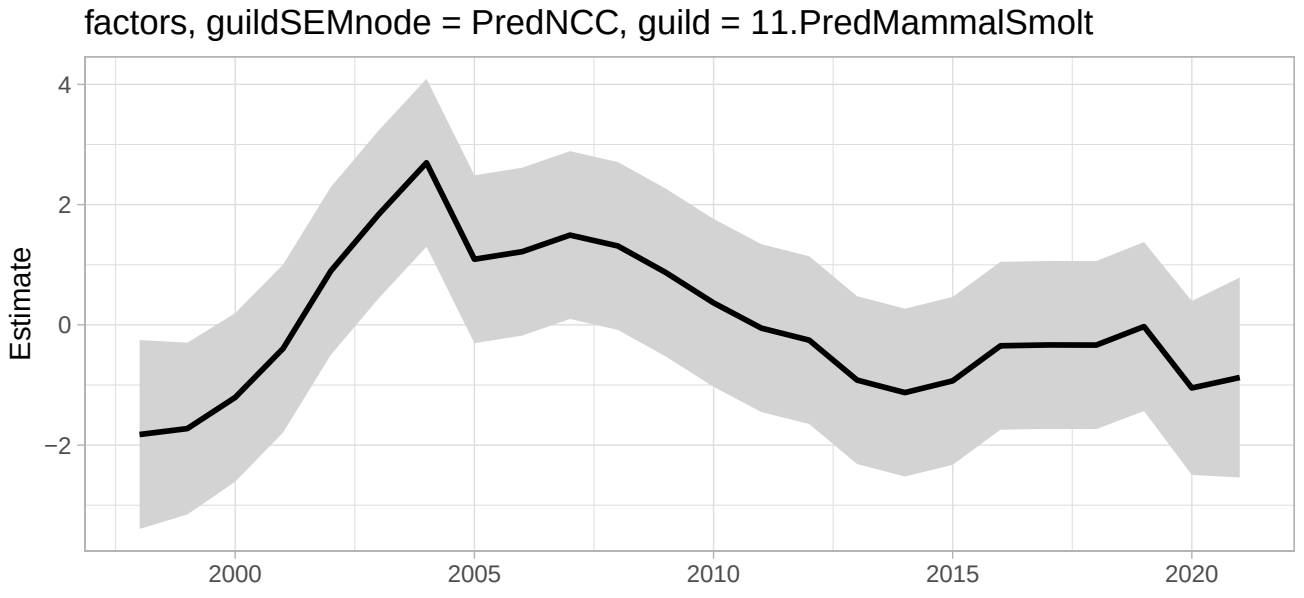


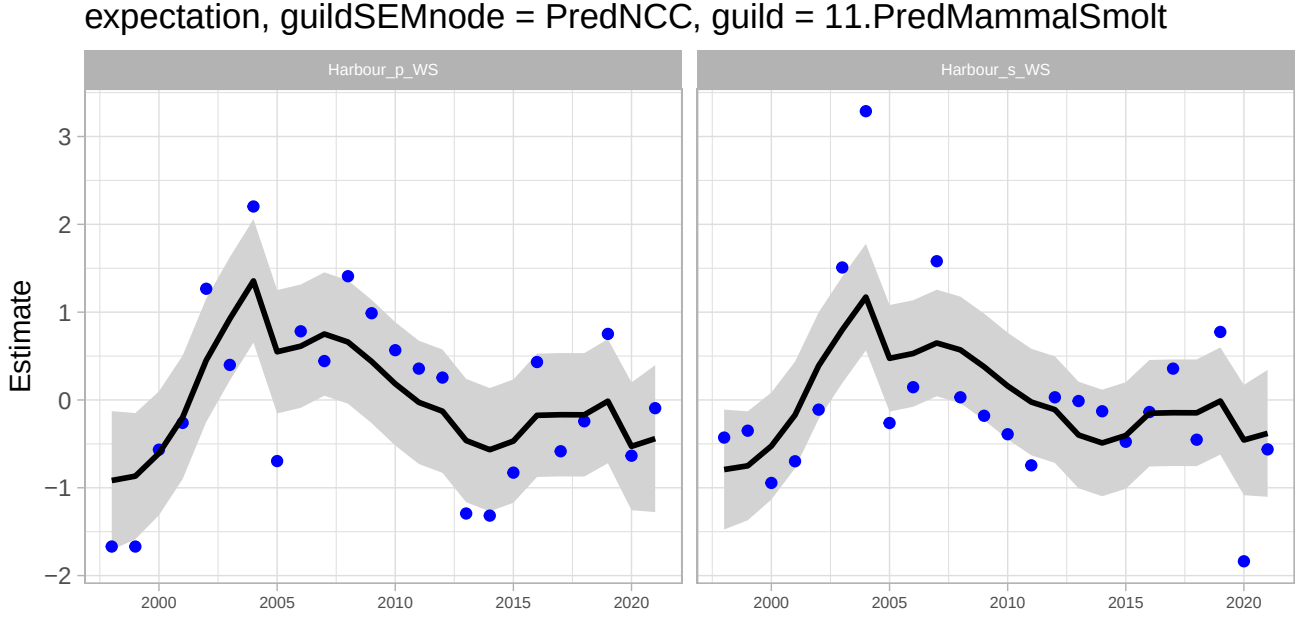
harbour_seal_DFO_Coastwide

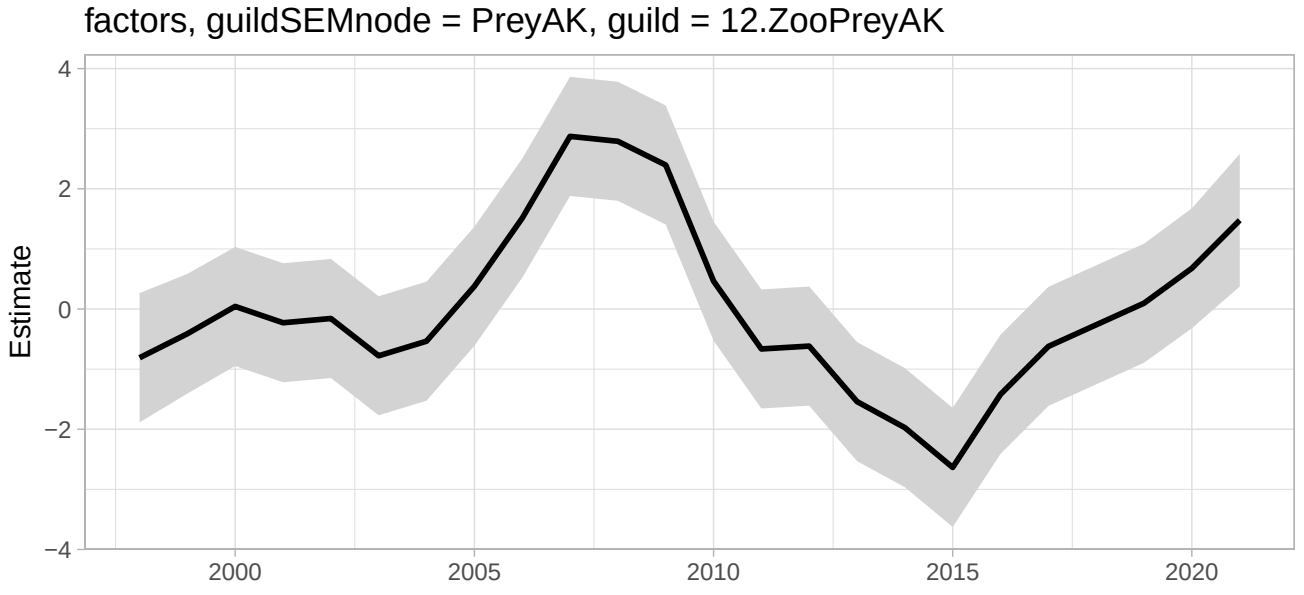


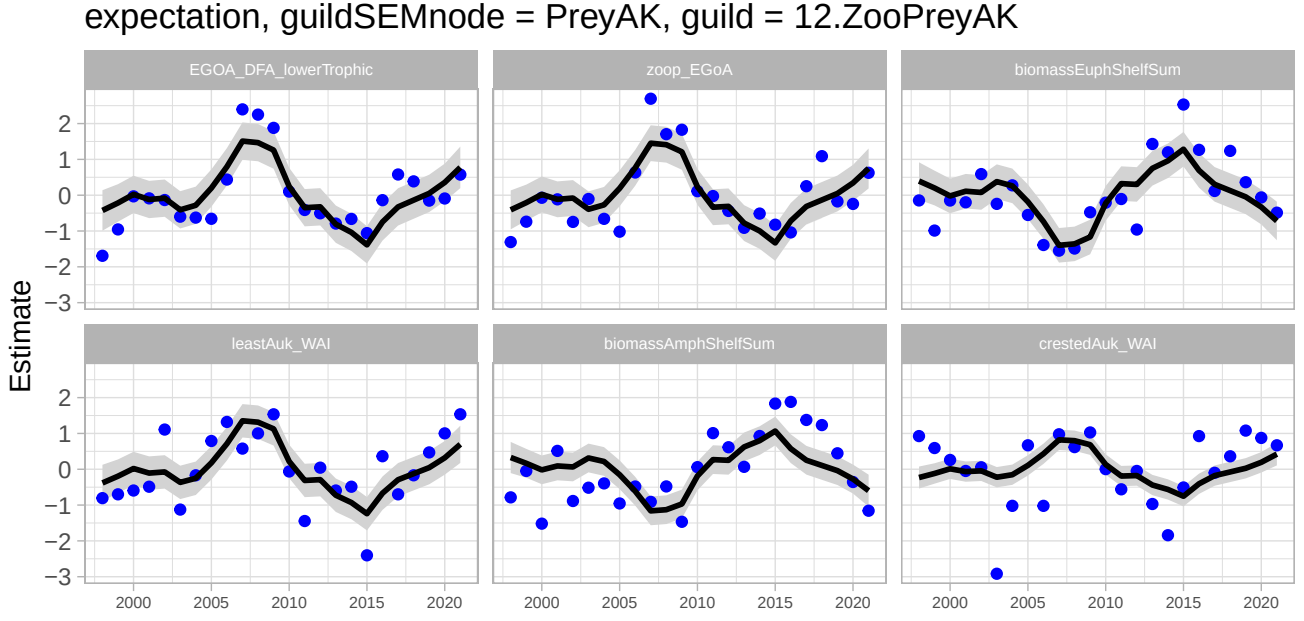

factors, guildSEMnode = PredAK, guild = 10.PredMammalNCC_CR
Estimate
5.0
2.5
0.0
−2.5
2000
2005
2010
2015
2020

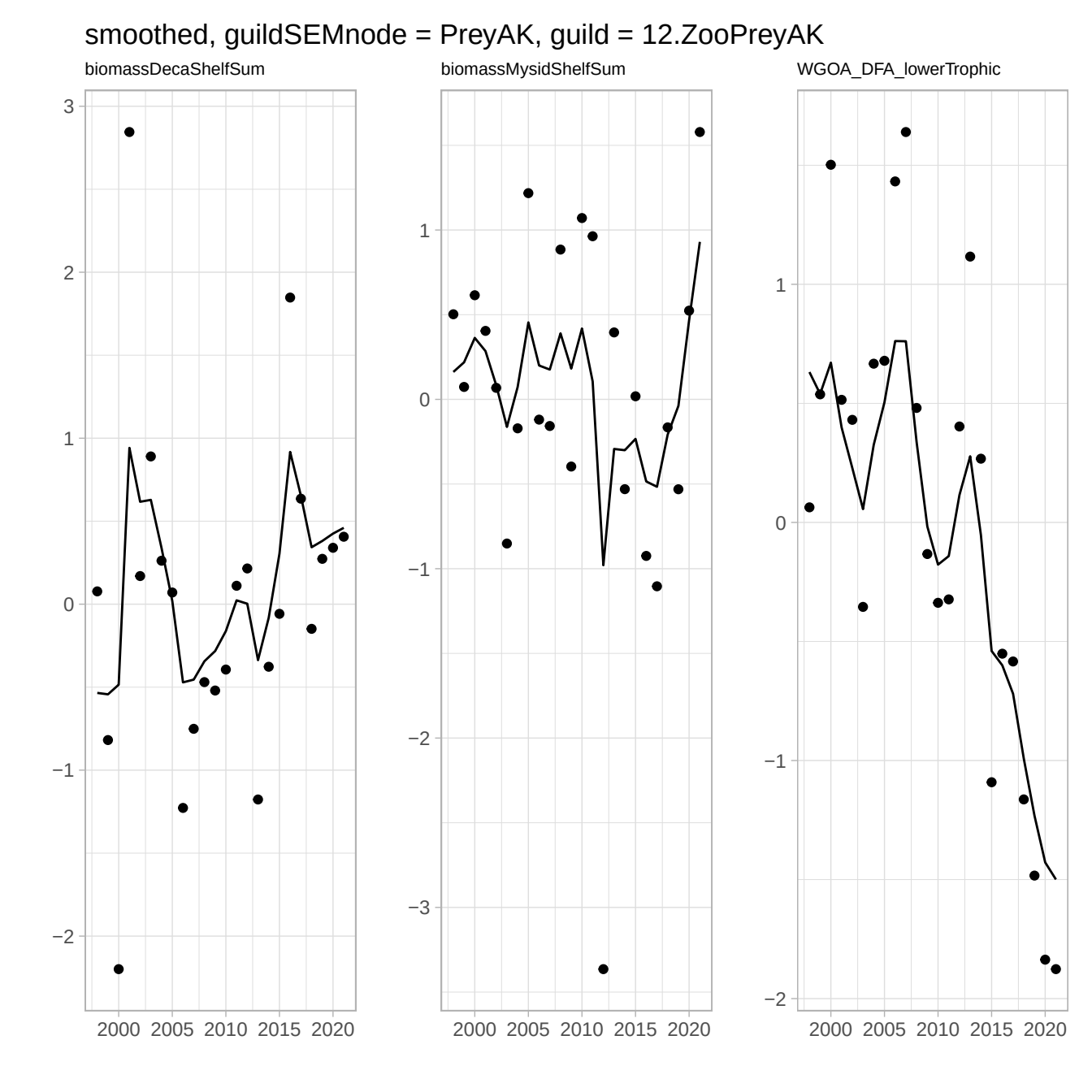

expectation, guildSEMnode = PredAK, guild = 10.PredMammalNCC_CR
AllSeaLionsEMB_adultyr
AllSeaLionsBonn_adultyr
Estimate
2
0
−2
2000
2005
2010
2015
2020

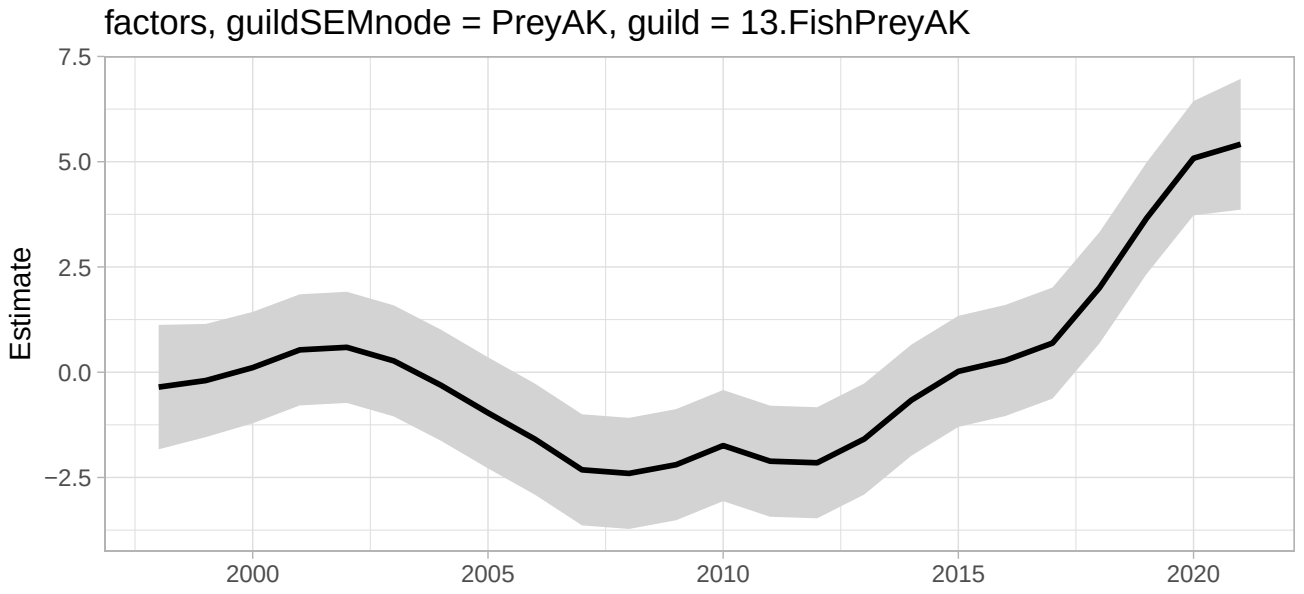

factors, guildSEMnode = PredNCC, guild = 11.PredMammalSmolt
Estimate
4
2
0
−2
2000
2005
2010
2015
2020

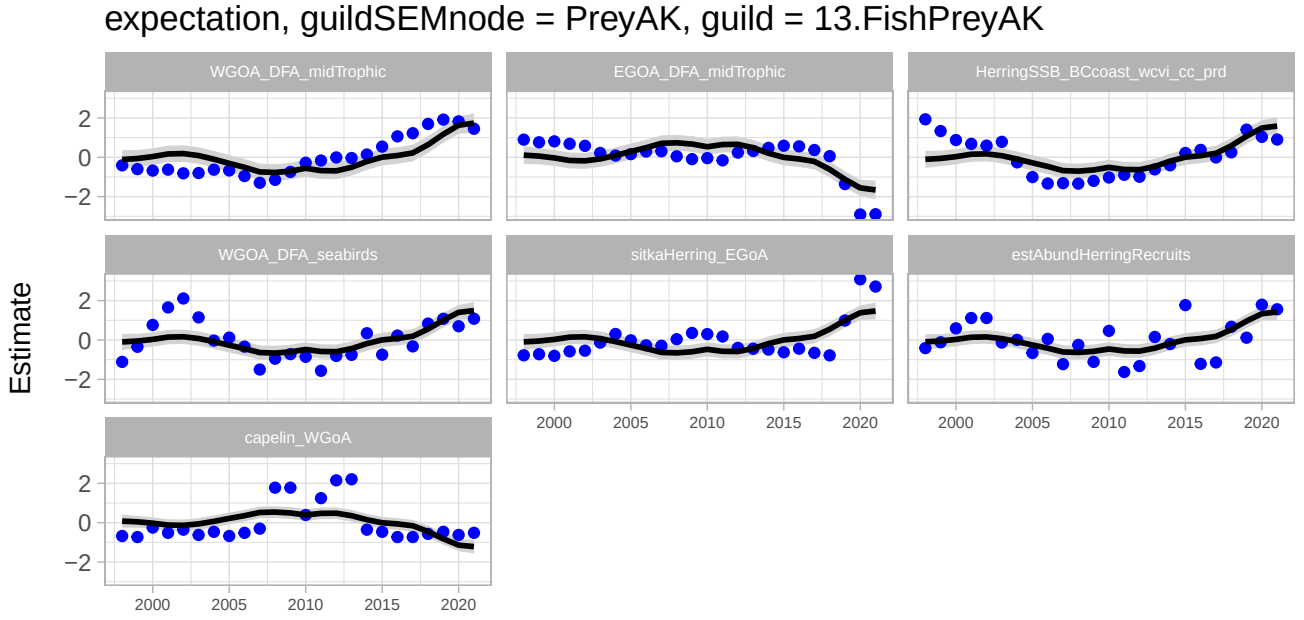

expectation, guildSEMnode = PredNCC, guild = 11.PredMammalSmolt
Harbour_p_WS
Harbour_s_WS
Estimate
3
2
1
0
−1
−2
2000
2005
2010
2015
2020

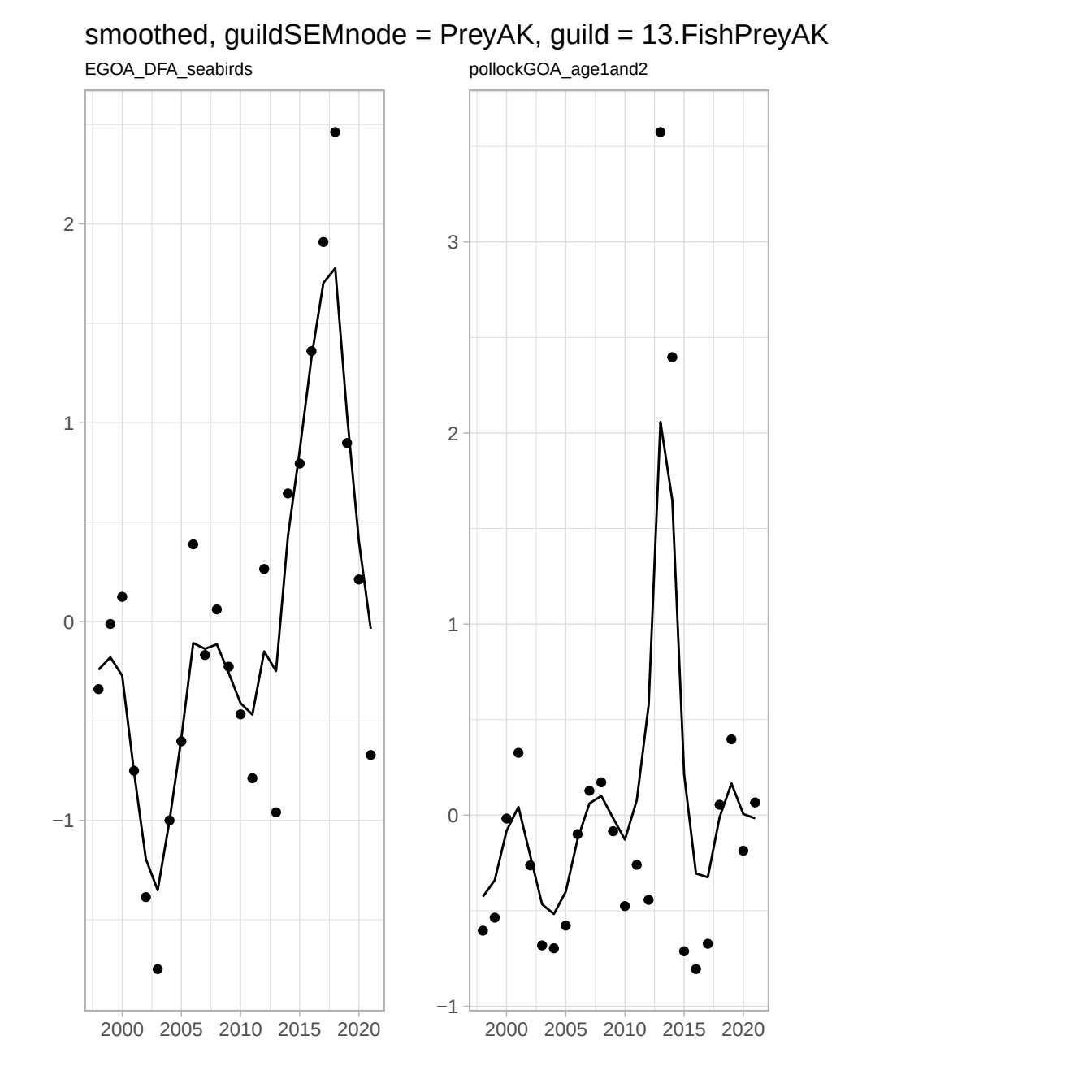

factors, guildSEMnode = PreyAK, guild = 12.ZooPreyAK
Estimate
4
2
0
−2
−4
2000
2005
2010
2015
2020

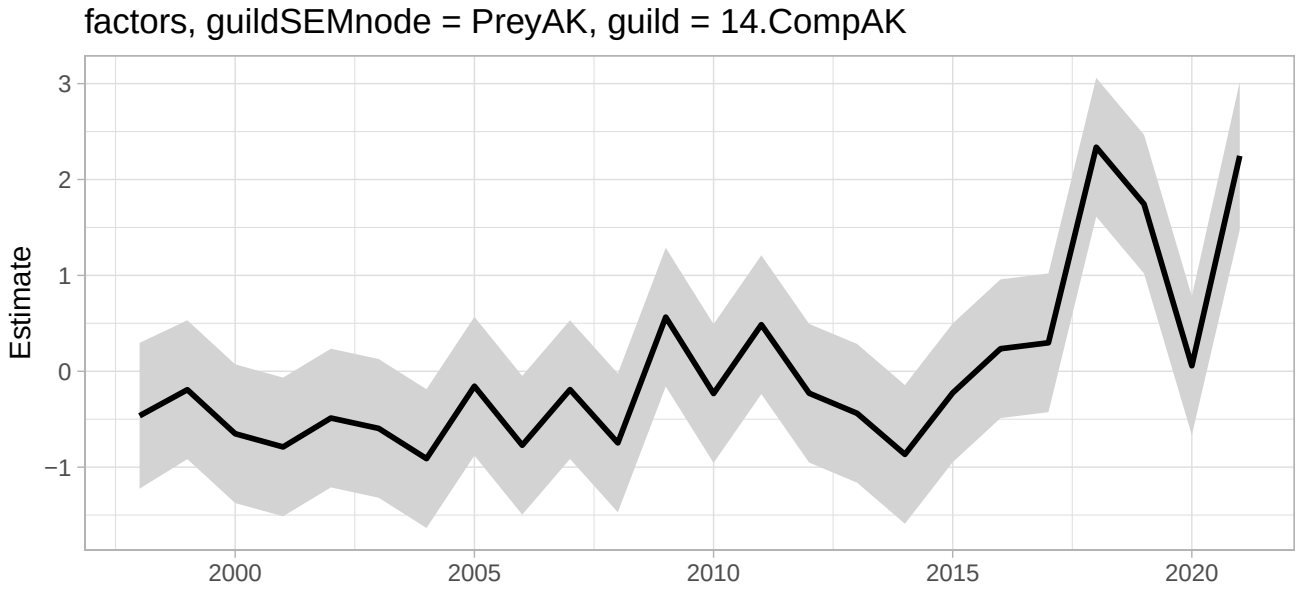

expectation, guildSEMnode = PreyAK, guild = 12.ZooPreyAK
Estimate
EGOA_DFA_lowerTrophic
zoop_EGoA
biomassEuphShelfSum
leastAuk_WAI
biomassAmphShelfSum
crestedAuk_WAI
2
1
0
−1
−2
−3
2000
2005
2010
2015
2020

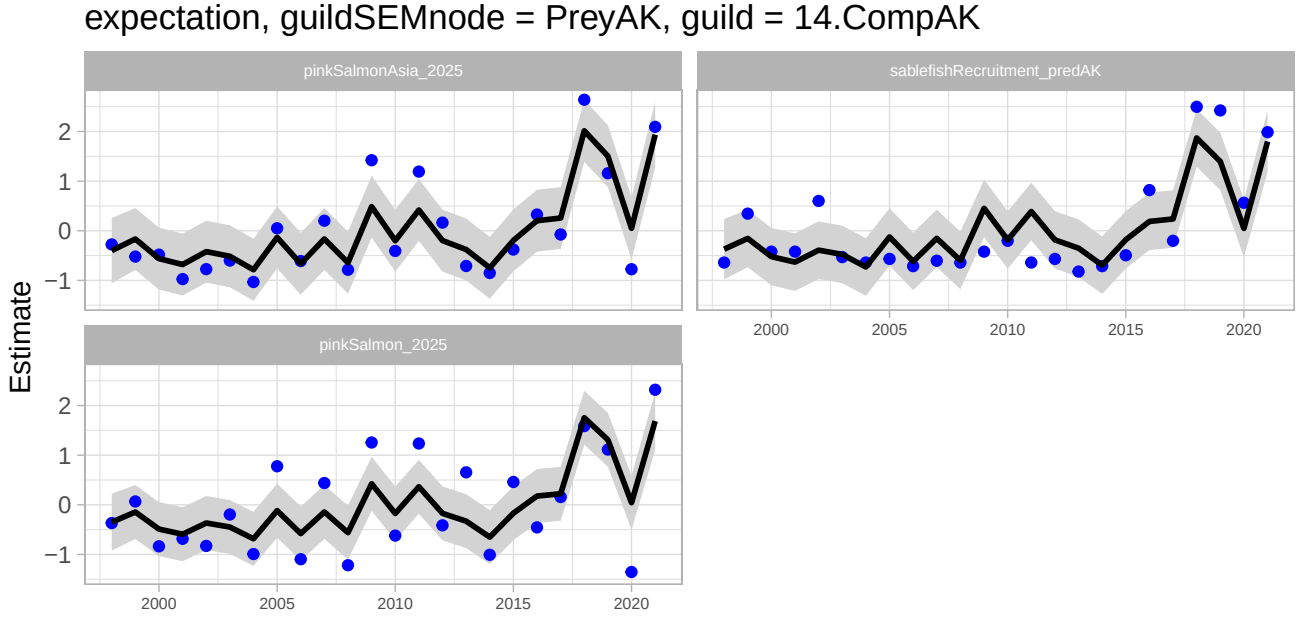

smoothed, guildSEMnode = PreyAK, guild = 12.ZooPreyAK
biomassDecaShelfSum
biomassMysidShelfSum
WGOA_DFA_lowerTrophic
3
2
1
0
−1
−2
1
0
−1
−2
−3
1
0
−1
−2
2000
2005
2010
2015
2020


factors, guildSEMnode = PreyAK, guild = 13.FishPreyAK
Estimate
7.5
5.0
2.5
0.0
−2.5
2000
2005
2010
2015
2020


expectation, guildSEMnode = PreyAK, guild = 13.FishPreyAK
Estimate
WGOA_DFA_midTrophic
EGOA_DFA_midTrophic
HerringSSB_BCcoast_wcvi_cc_prd
WGOA_DFA_seabirds
sitkaHerring_EGoA
estAbundHerringRecruits
capelin_WGoA
2
0
−2
2000
2005
2010
2015
2020

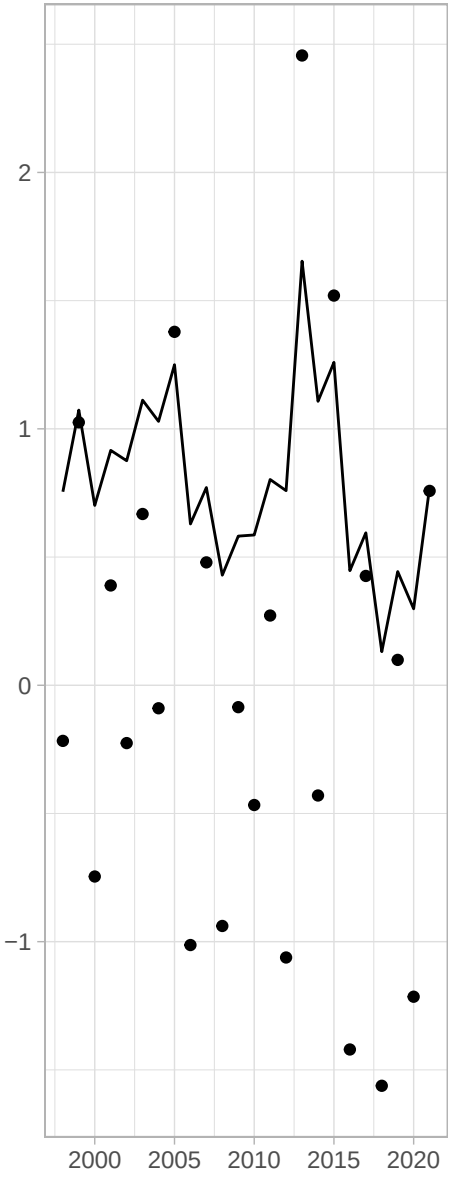

smoothed, guildSEMnode = PreyAK, guild = 13.FishPreyAK
EGOA_DFA_seabirds
pollockGOA_age1and2
2
1
0
−1
3
2
1
0
−1
2000
2005
2010
2015
2020

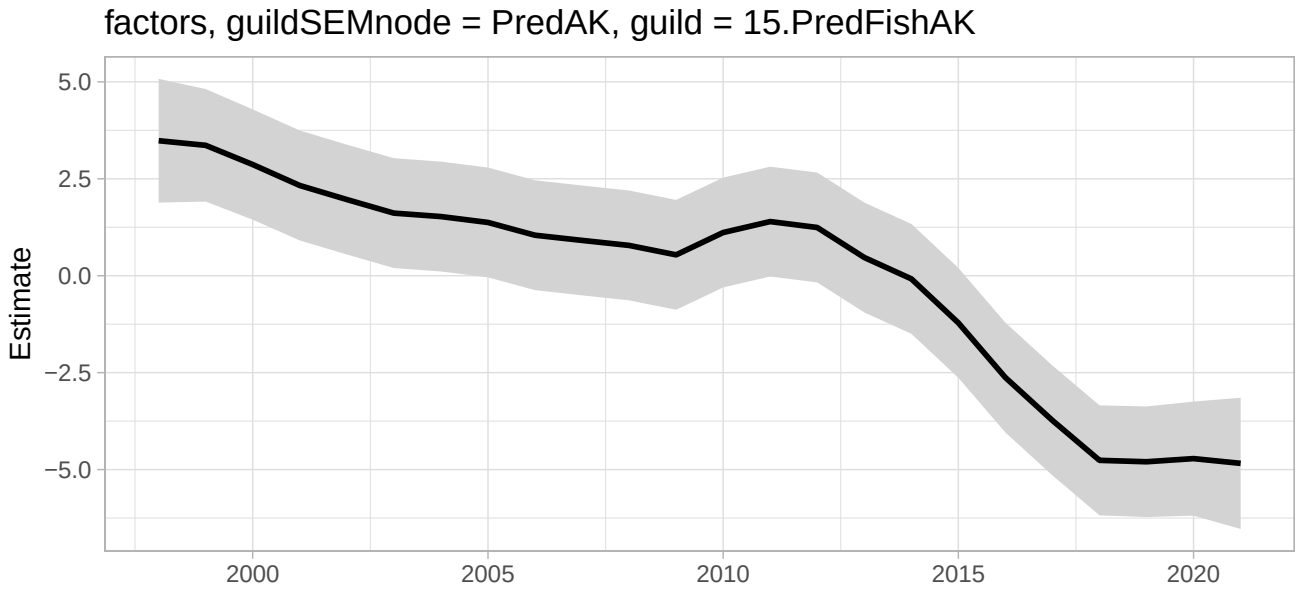

factors, guildSEMnode = PreyAK, guild = 14.CompAK
Estimate
3
2
1
0
−1
2000
2005
2010
2015
2020

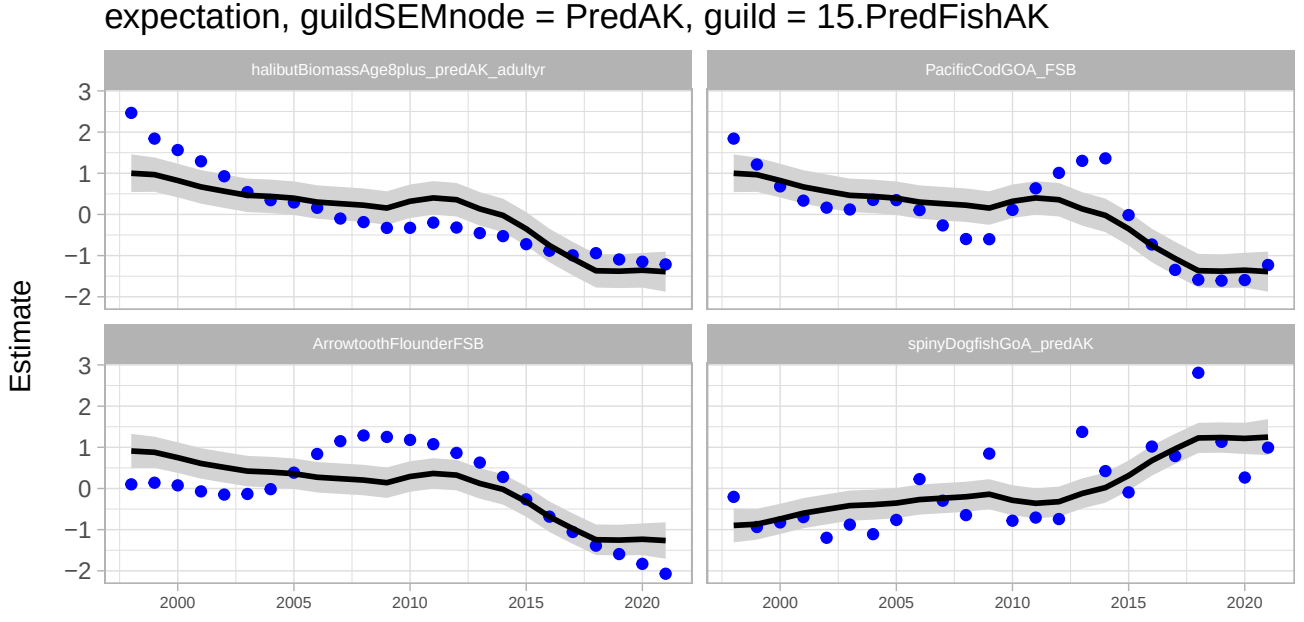

expectation, guildSEMnode = PreyAK, guild = 14.CompAK
Estimate
pinkSalmonAsia_2025
sablefishRecruitment_predAK
pinkSalmon_2025
2
1
0
−1
2000
2005
2010
2015
2020

smoothed, guildSEMnode = PreyAK, guild = 14.CompAK

pinkSalmonNorthAmerica_2025

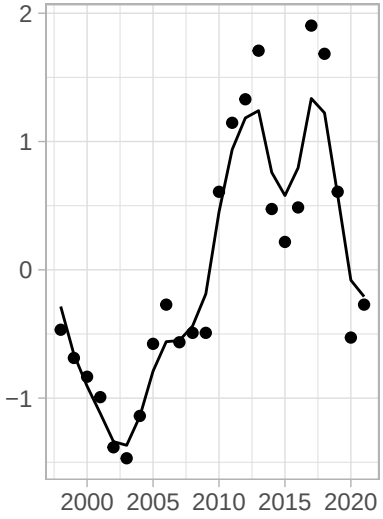



factors, guildSEMnode = PredAK, guild = 15.PredFishAK
Estimate
5.0
2.5
0.0
−2.5
−5.0
2000
2005
2010
2015
2020

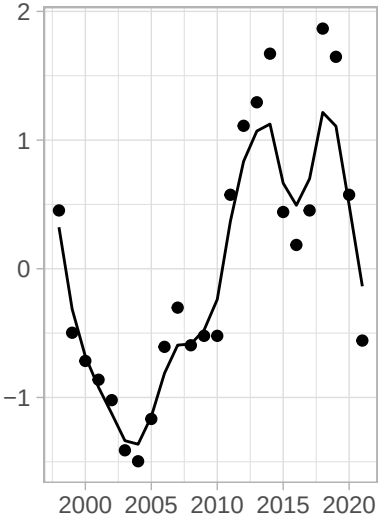

expectation, guildSEMnode = PredAK, guild = 15.PredFishAK
Estimate
halibutBiomassAge8plus_predAK_adultyr
PacificCodGOA_FSB
ArrowtoothFlounderFSB
spinyDogfishGoA_predAK
3
2
1
0
−1
−2
2000
2005
2010
2015
2020

smoothed, guildSEMnode = PredAK, guild = 15.PredFishAK

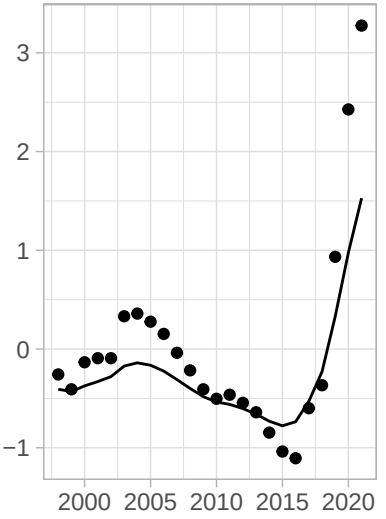
pollockGOA_SSB



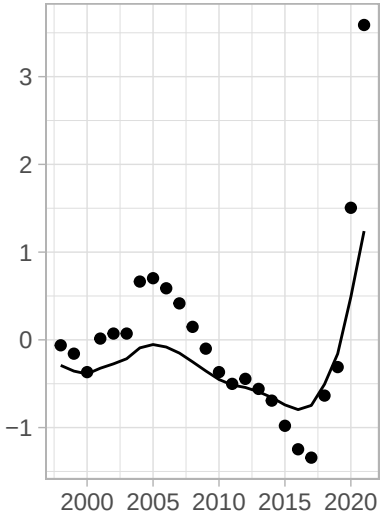
pollockGOA_SSB_lag1

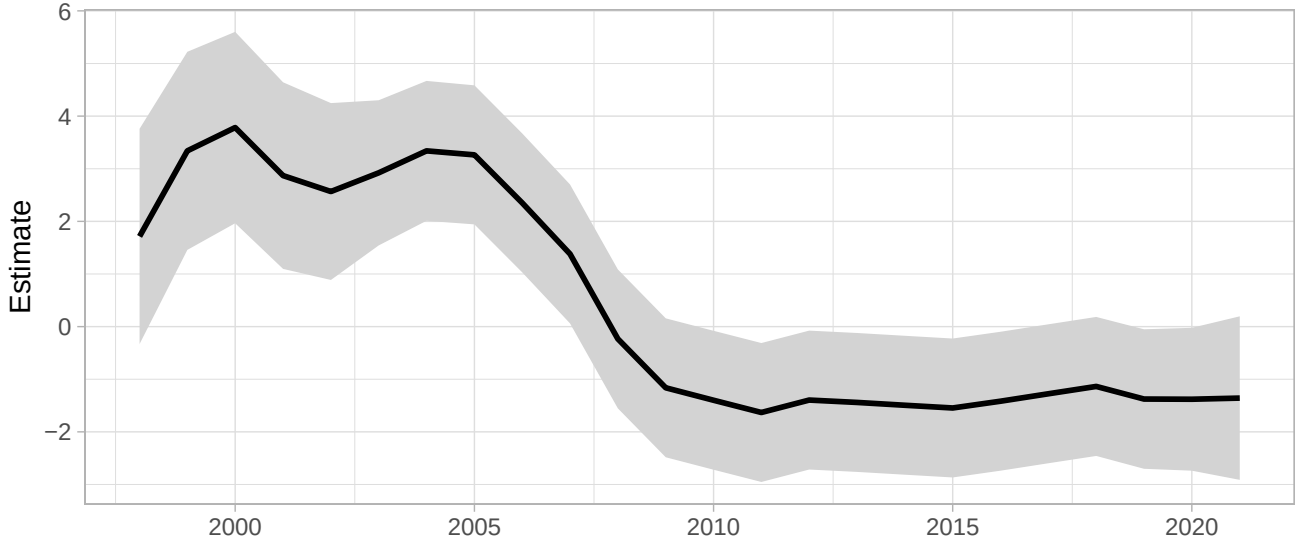


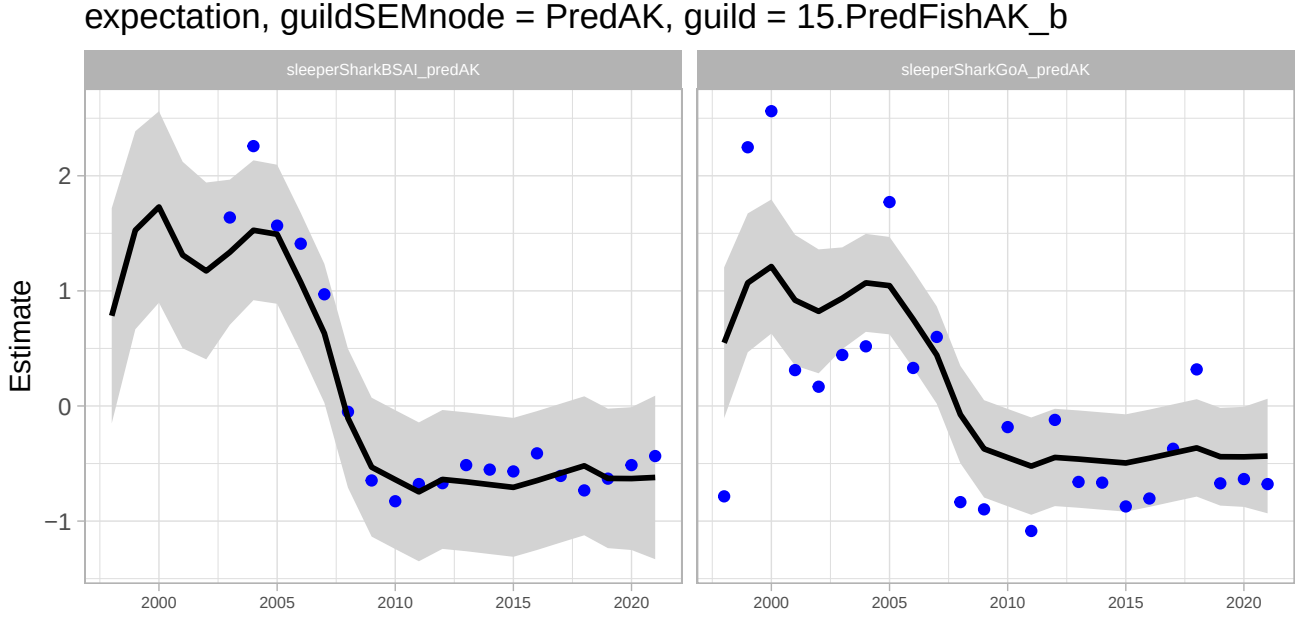
sablefishAK_biomass3plus

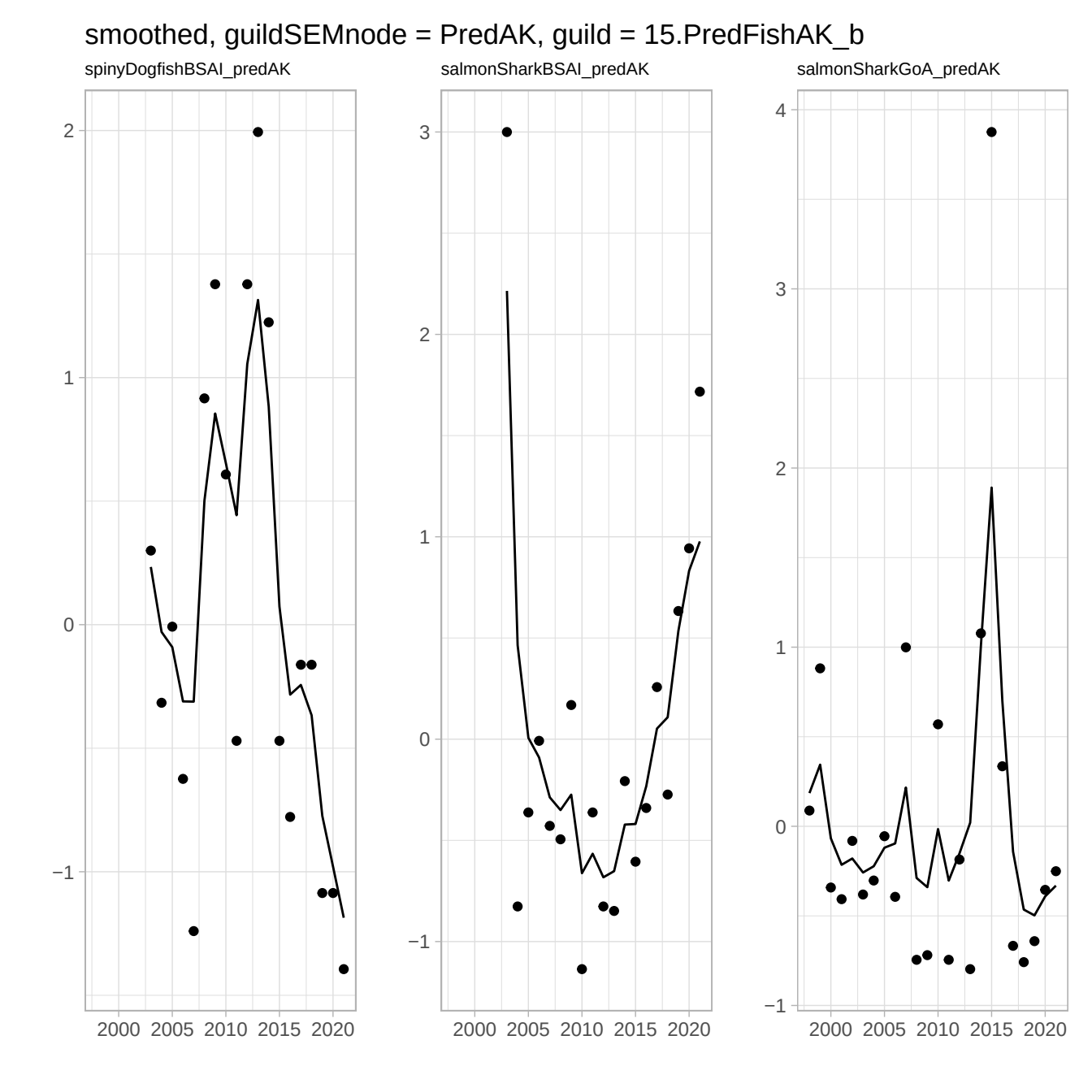


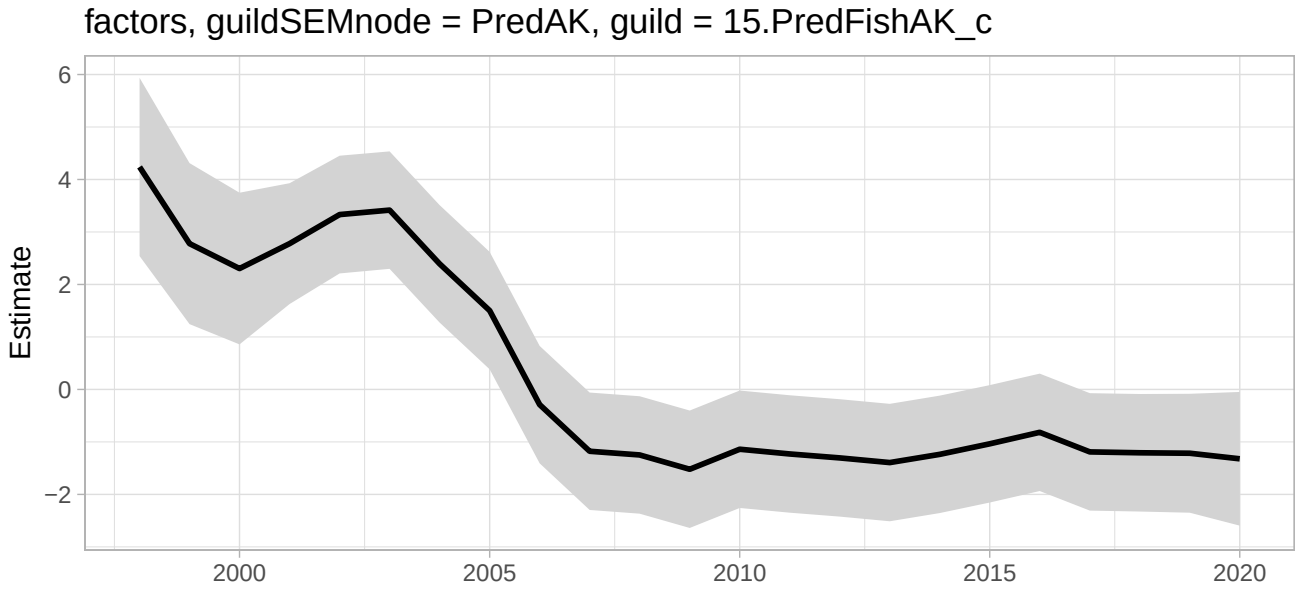
sablefishAK_biomass4plus

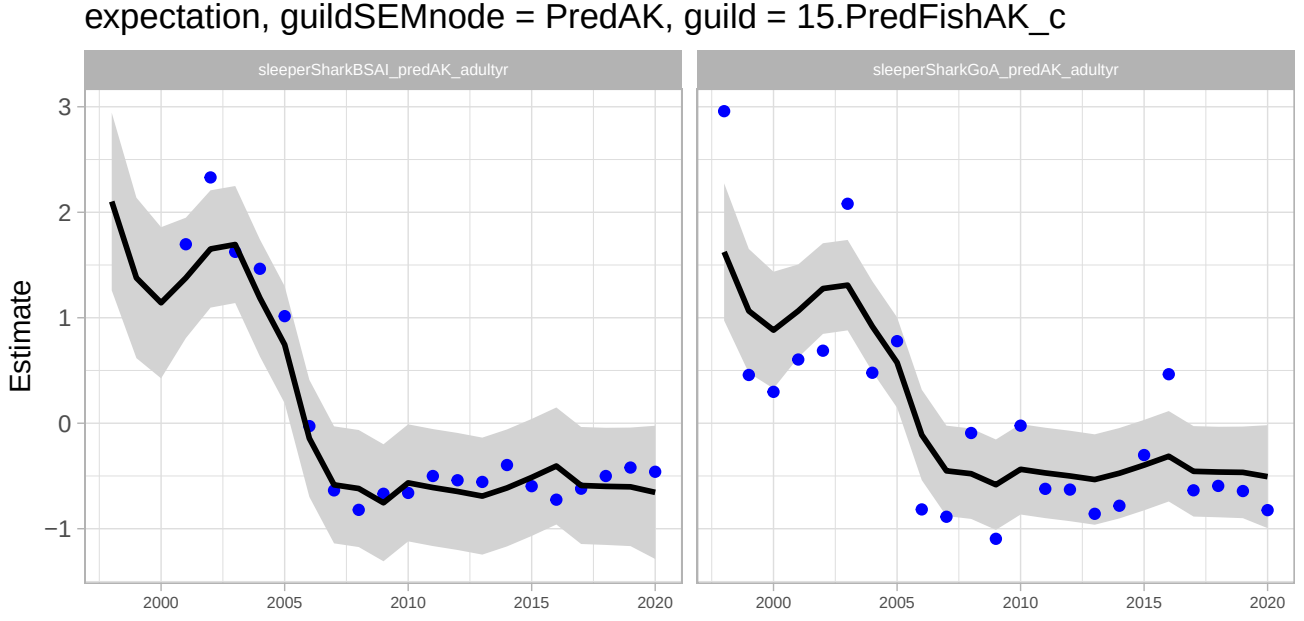


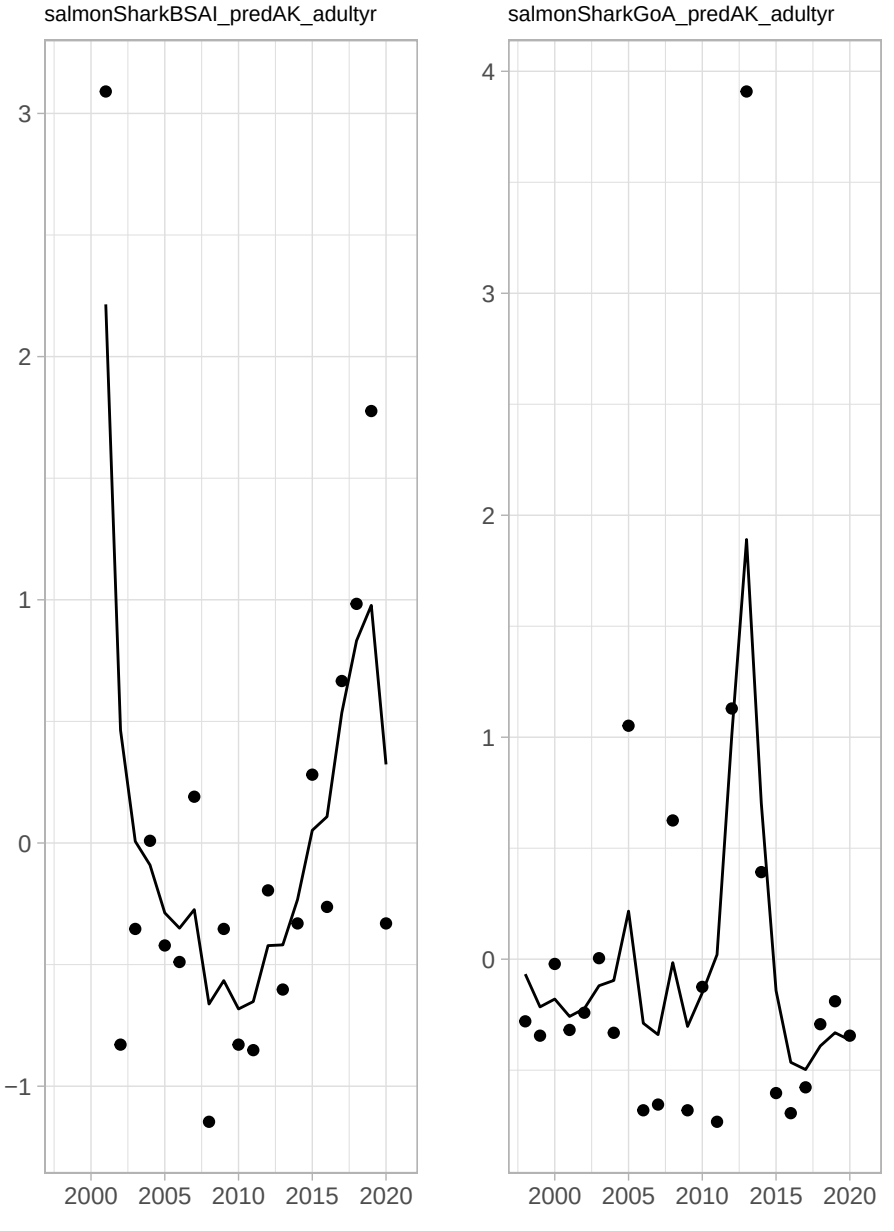

factors, guildSEMnode = PredAK, guild = 15.PredFishAK_b
Estimate
6
4
2
0
−2
2000
2005
2010
2015
2020

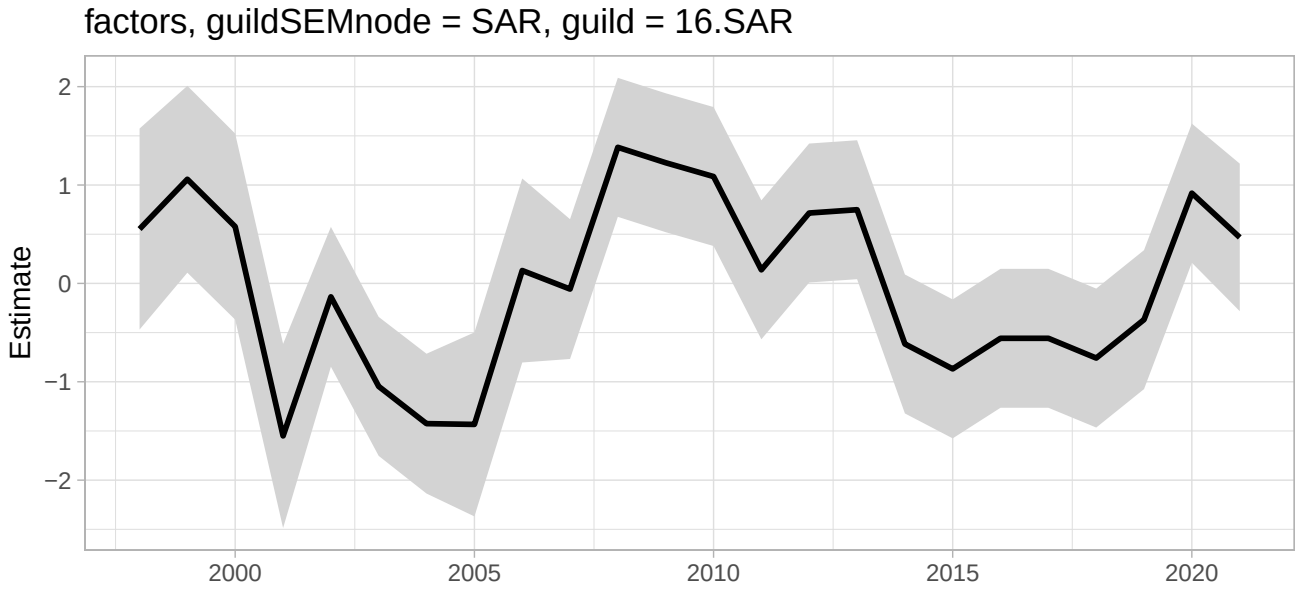

expectation, guildSEMnode = PredAK, guild = 15.PredFishAK_b
sleeperSharkBSAI_predAK
sleeperSharkGoA_predAK
Estimate
2
1
0
−1
2000
2005
2010
2015
2020

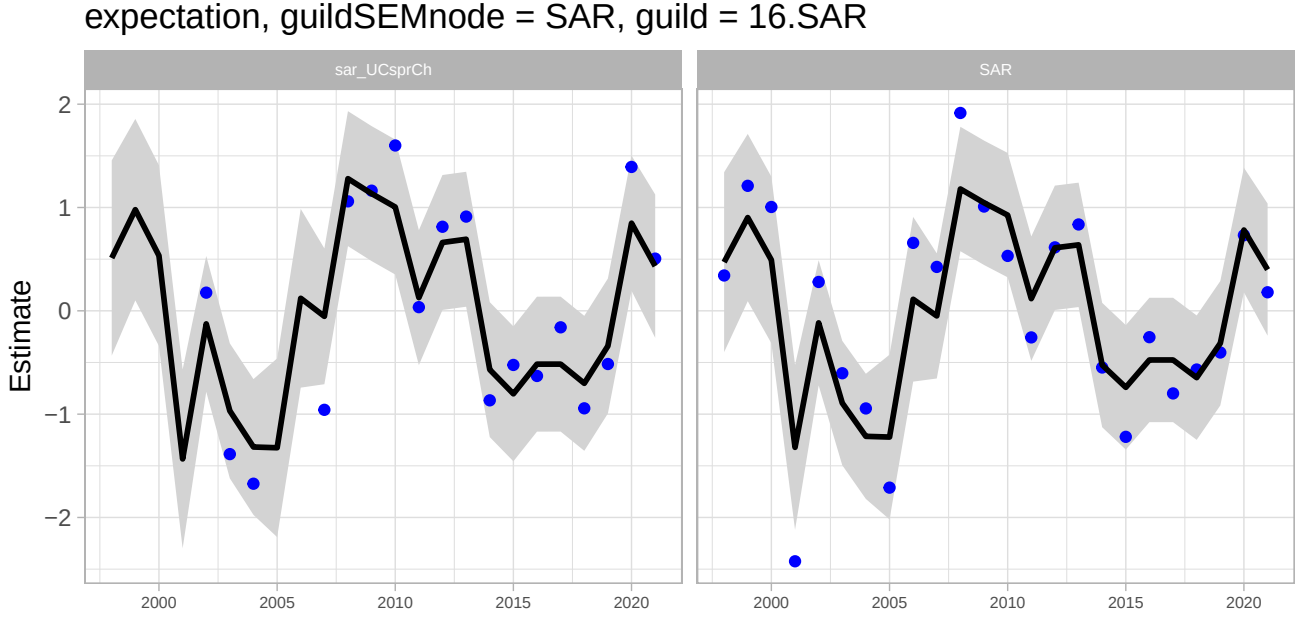

smoothed, guildSEMnode = PredAK, guild = 15.PredFishAK_b
spinyDogfishBSAI_predAK
salmonSharkBSAI_predAK
salmonSharkGoA_predAK
2
1
0
−1
3
2
1
0
−1
4
3
2
1
0
−1
2000
2005
2010
2015
2020

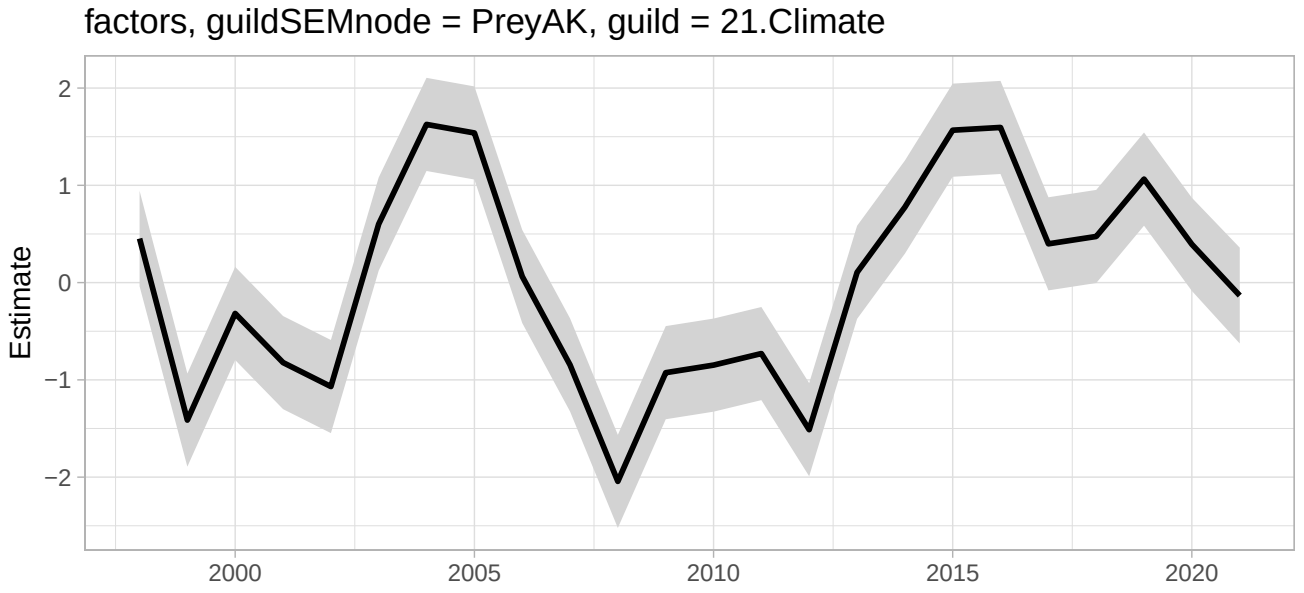

factors, guildSEMnode = PredAK, guild = 15.PredFishAK_c
Estimate
6
4
2
0
−2
2000
2005
2010
2015
2020

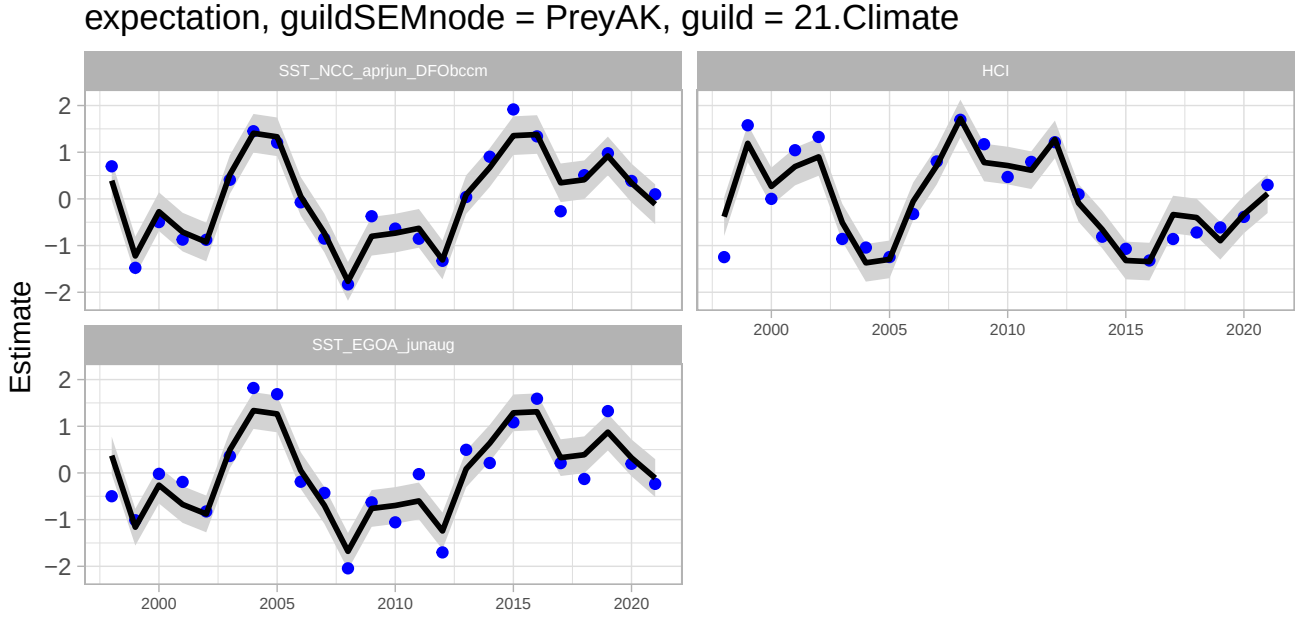

expectation, guildSEMnode = PredAK, guild = 15.PredFishAK_c
sleeperSharkBSAI_predAK_adultyr
sleeperSharkGoA_predAK_adultyr
Estimate
3
2
1
0
−1
2000
2005
2010
2015
2020

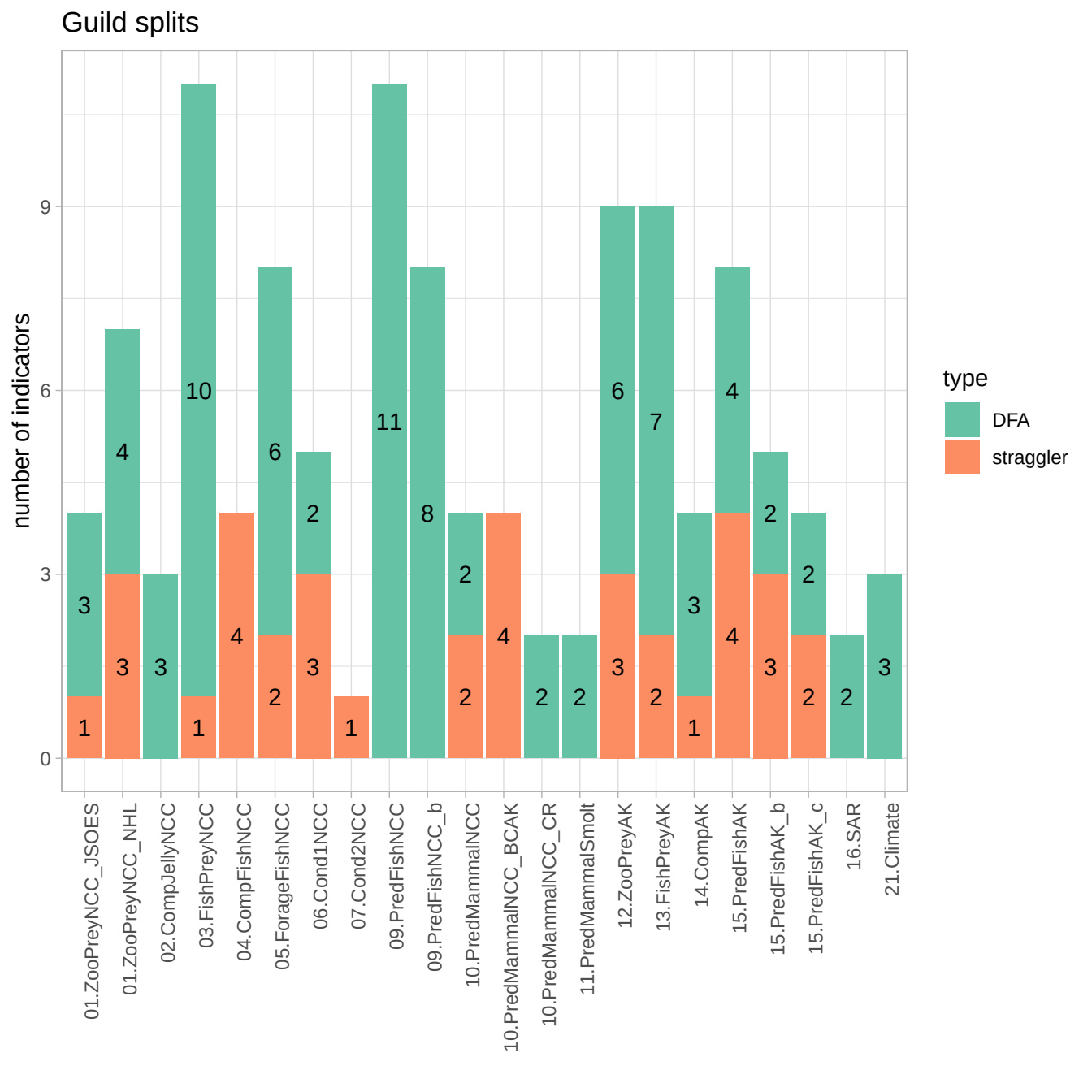

smoothed, guildSEMnode = PredAK, guild = 15.PredFishAK_c
salmonSharkBSAI_predAK_adultyr
salmonSharkGoA_predAK_adultyr
3
2
1
0
−1
4
3
2
1
0
2000
2005
2010
2015
2020

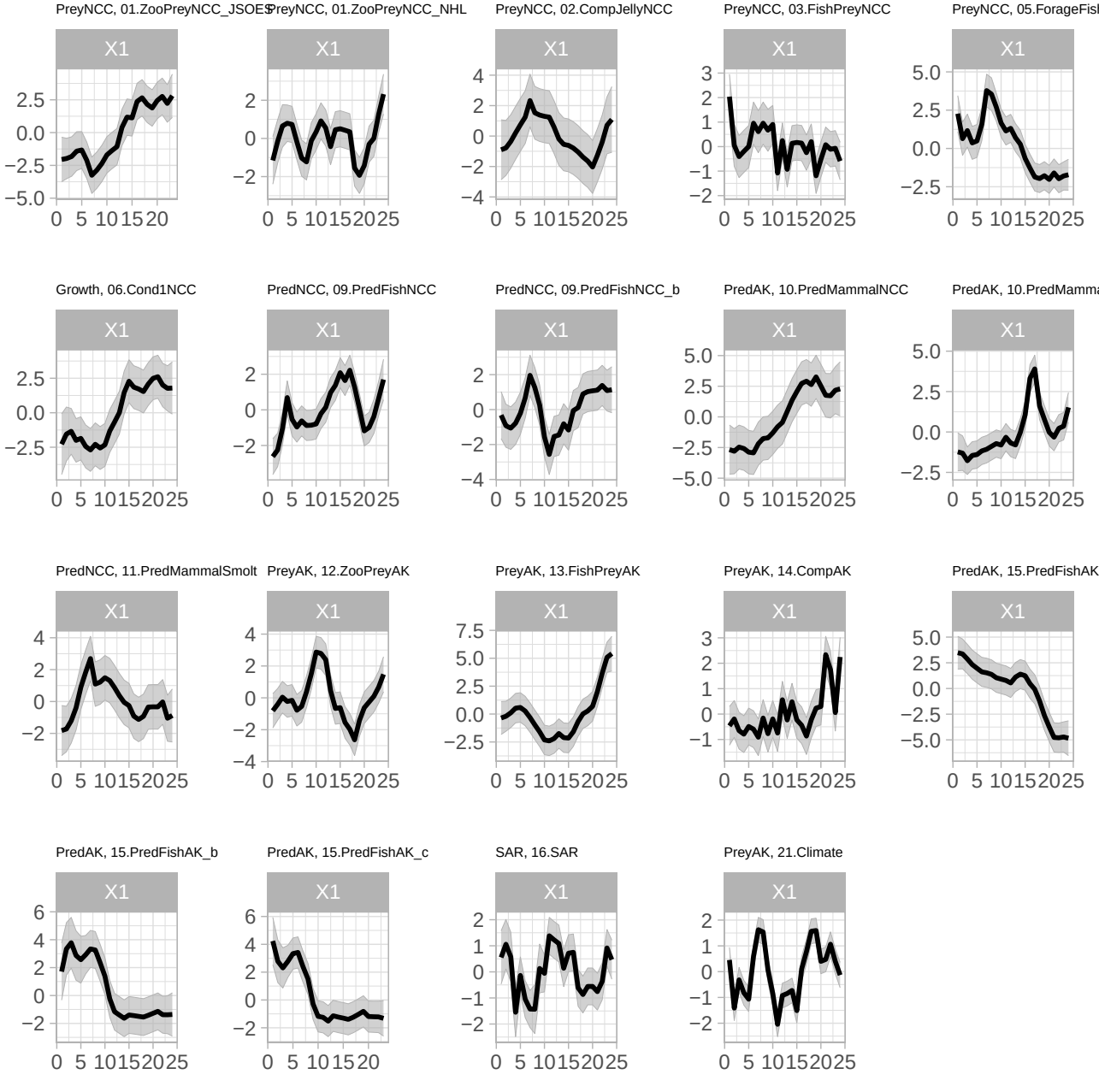

factors, guildSEMnode = SAR, guild = 16.SAR
Estimate
2
1
0
−1
−2
2000
2005
2010
2015
2020


expectation, guildSEMnode = SAR, guild = 16.SAR
sar_UCsprCh
SAR
Estimate
2
1
0
−1
−2
2000
2005
2010
2015
2020


factors, guildSEMnode = PreyAK, guild = 21.Climate
Estimate
2
1
0
−1
−2
2000
2005
2010
2015
2020


expectation, guildSEMnode = PreyAK, guild = 21.Climate
SST_NCC_aprjun_DFObccm
HCI
SST_EGOA_junaug
Estimate
2
1
0
−1
−2
2000
2005
2010
2015
2020


Guild splits
number of indicators
0
3
6
9
type
DFA
straggler
3
1
4
3
3
10
1
4
6
2
2
3
1
11
8
2
2
4
2
2
6
3
7
2
3
1
4
4
2
3
3
2
2
2
3
01.ZooPreyNCC_JSOES
01.ZooPreyNCC_NHL
02.CompJellyNCC
03.FishPreyNCC
04.CompFishNCC
05.ForageFishNCC
06.Cond1NCC
07.Cond2NCC
09.PredFishNCC
09.PredFishNCC_b
10.PredMammalNCC
10.PredMammalNCC_BCAK
10.PredMammalNCC_CR
11.PredMammalSmolt
12.ZooPreyAK
13.FishPreyAK
14.CompAK
15.PredFishAK
15.PredFishAK_b
15.PredFishAK_c
16.SAR
21.Climate


PreyNCC, 01.ZooPreyNCC_JSOES
PreyNCC, 01.ZooPreyNCC_NHL
PreyNCC, 02.CompJellyNCC
PreyNCC, 03.FishPreyNCC
PreyNCC, 05.ForageFish
Growth, 06.Cond1NCC
PredNCC, 09.PredFishNCC
PredNCC, 09.PredFishNCC_b
PredAK, 10.PredMammalNCC
PredAK, 10.PredMamma
PredNCC, 11.PredMammalSmolt
PreyAK, 12.ZooPreyAK
PreyAK, 13.FishPreyAK
PreyAK, 14.CompAK
PredAK, 15.PredFishAK
PredAK, 15.PredFishAK_b
PredAK, 15.PredFishAK_c
SAR, 16.SAR
PreyAK, 21.Climate
X1